

Mathematics

Basic 8

Complete BECE Preparation
Examination Papers & Marking Schemes

Beacon Educational Consult

2026 Edition

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Preface

This comprehensive Mathematics textbook for Basic 8 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

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Chapter Examinations (17 Papers)

Chapters 1-17: Mock Examinations with Answer Keys

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BEACON EDUCATIONAL SERIES

Mathematics for Basic 8

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 8 (JHS 2) Students and Teachers

Beacon Educational Consult
2026 Edition

Mathematics for Basic 8
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 8 (JHS 2) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY

Chapter 1: Ch10

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Angles on a straight line add up to
 - A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
2. The sum of angles at a point is
 - A. 180°
 - B. 90°
 - C. 270°
 - D. 360°
3. Two straight lines intersect and one of the angles formed is 70° . The vertically opposite angle is
 - A. 70°
 - B. 110°
 - C. 90°
 - D. 180°
4. Two angles on a straight line are x° and 115° . The value of x is
 - A. 55°
 - B. 75°
 - C. 65°
 - D. 115°
5. Angles around a point are 120° , 85° , 90° and x° . The value of x is
 - A. 75°
 - B. 65°
 - C. 55°
 - D. 95°
6. The supplement of an angle is three times the angle. The angle is
 - A. 60°
 - B. 30°
 - C. 45°
 - D. 90°
7. The complement of 38° is
 - A. 52°
 - B. 42°
 - C. 62°
 - D. 142°
8. Two vertically opposite angles are $(2x + 10)^\circ$ and 70° . The value of x is
 - A. 40
 - B. 25
 - C. 35
 - D. 30
9. When a transversal cuts two parallel lines, the angles in matching positions are called
 - A. alternate angles
 - B. corresponding angles
 - C. co-interior angles
 - D. vertically opposite angles
10. A transversal cuts two parallel lines. If one of a pair of corresponding angles is 72° , the other is
 - A. 72°
 - B. 108°
 - C. 18°
 - D. 90°
11. A transversal cuts two parallel lines. If one of a pair of alternate angles is 55° , the other is
 - A. 125°
 - B. 35°
 - C. 45°
 - D. 55°
12. When a transversal cuts two parallel lines, a pair of co-interior angles is
 - A. equal
 - B. supplementary
 - C. complementary
 - D. vertically opposite

- 13.** A transversal cuts two parallel lines. If one of a pair of co-interior angles is 124° , the other is
- A. 124°
 - B. 66°
 - C. 56°
 - D. 46°
- 15.** A transversal cuts two parallel lines. If one of the eight angles formed is 118° , the number of the eight angles equal to 118° is
- A. 2
 - B. 8
 - C. 6
 - D. 4
- 17.** A transversal cuts two parallel lines. If the difference between a pair of co-interior angles is 40° , the smaller angle is
- A. 70°
 - B. 110°
 - C. 80°
 - D. 40°
- 19.** A transversal cuts two parallel lines. If one angle is 37° , the supplement of its corresponding angle is
- A. 37°
 - B. 53°
 - C. 127°
 - D. 143°
- 21.** The sum of the angles in a triangle is
- A. 360°
 - B. 180°
 - C. 90°
 - D. 270°
- 23.** In triangle PQR, $\angle P = 2x^\circ$, $\angle Q = 3x^\circ$ and $\angle R = 4x^\circ$. The value of x is
- A. 20
 - B. 18
 - C. 25
 - D. 30
- 25.** Two interior angles of a triangle are 65° and 48° . The exterior angle at the third vertex is
- A. 67°
 - B. 113°
 - C. 123°
- 14.** A transversal cuts two parallel lines. One of a pair of co-interior angles is three times the other. The smaller angle is
- A. 45°
 - B. 60°
 - C. 135°
 - D. 30°
- 16.** A transversal cuts two parallel lines. Two angles on a straight line are y° and 132° . If y° is alternate to x° , then $x =$
- A. 132°
 - B. 58°
 - C. 48°
 - D. 42°
- 18.** A transversal cuts two parallel lines. One angle is $2x^\circ$ and its alternate angle is $(x + 50)^\circ$. The value of x is
- A. 25
 - B. 50
 - C. 100
 - D. 75
- 20.** Which of the following shows that two lines cut by a transversal are parallel?
- A. Co-interior angles are equal
 - B. Alternate angles are supplementary
 - C. Corresponding angles are equal
 - D. Vertically opposite angles are equal
- 22.** In triangle ABC, $\angle A = 50^\circ$ and $\angle B = 60^\circ$. $\angle C =$
- A. 80°
 - B. 60°
 - C. 110°
 - D. 70°
- 24.** The exterior angle of a triangle is equal to
- A. 180°
 - B. the third interior angle
 - C. the sum of the two interior opposite angles
 - D. 360°

D. 97°

- 27.** The sum of the interior angles of a hexagon is
A. 540°
B. 900°
C. 720°
D. 630°
- 29.** The sum of the interior angles of a pentagon is
A. 720°
B. 540°
C. 450°
D. 360°
- 31.** Each exterior angle of a regular octagon is
A. 135°
B. 60°
C. 36°
D. 45°
- 33.** Each interior angle of a regular polygon is 144° . The polygon is a
A. decagon
B. nonagon
C. octagon
D. dodecagon
- 35.** In triangle ABC, $AB = AC$ and $\angle B = 68^\circ$. $\angle A =$
A. 68°
B. 112°
C. 22°
D. 44°
- 37.** The angles of a triangle are in the ratio $2 : 3 : 4$. The greatest angle is
A. 60°
B. 90°
C. 80°
D. 40°
- 39.** The sum of the interior angles of a polygon is 1440° . The number of sides of the polygon is
A. 8
B. 12
C. 9
D. 10
- 26.** The sum of the interior angles of a polygon of n sides is
A. $(n - 2) \times 180^\circ$
B. $n \times 180^\circ$
C. $(n + 2) \times 180^\circ$
D. 360°
- 28.** Each interior angle of a regular hexagon is
A. 108°
B. 135°
C. 60°
D. 120°
- 30.** The sum of the exterior angles of any convex polygon is
A. 360°
B. 180°
C. $(n - 2) \times 180^\circ$
D. 270°
- 32.** A regular polygon has an interior angle of 135° . The number of sides of the polygon is
A. 6
B. 10
C. 8
D. 9
- 34.** The angles of a triangle are x° , $(x + 20)^\circ$ and $(x + 40)^\circ$. The smallest angle is
A. 60°
B. 40°
C. 80°
D. 50°
- 36.** In an isosceles triangle, the vertical angle is 40° . Each base angle is
A. 140°
B. 70°
C. 40°
D. 100°
- 38.** A transversal cuts two parallel lines. One angle is three times its adjacent angle on the same straight line. The smaller angle is
A. 45°
B. 60°
C. 135°
D. 30°

- 40.** In a regular polygon, the ratio of an interior angle to its exterior angle is 4 : 1. The number of sides of the polygon is
- A. 8
 - B. 10
 - C. 12
 - D. 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Angle facts.

Question 2

(a) Transversals.

Question 3

(a) Angles in triangles.

Question 4

(a) Angles in polygons.

Question 5

(a) Applications.

Question 6

(a) Reasoning and deduction.

Chapter 2: Ch11

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following angles cannot be constructed using a pair of compasses and a ruler only?
 - A. 75°
 - B. 80°
 - C. 105°
 - D. 135°
3. To bisect an angle means to
 - A. divide it into three equal angles
 - B. double the angle
 - C. divide it into two equal angles
 - D. make it a right angle
5. Bisecting an angle of 150° gives two angles each of
 - A. 75°
 - B. 50°
 - C. 80°
 - D. 60°
7. To bisect an angle, arcs are drawn from the two points where the first arc cuts the arms of the angle. These two arcs must have
 - A. different radii
 - B. radii equal to the lengths of the arms
 - C. equal radii
 - D. radii equal to the size of the angle
9. The instrument used to check the size of a constructed angle of 105° is
 - A. a ruler
 - B. a pair of set squares only
 - C. a divider
 - D. a protractor
11. To construct an equilateral triangle, it is enough to know
 - A. one angle
 - B. two sides
 - C. two angles
2. The first step in constructing an angle of 60° with a pair of compasses is to
 - A. draw a ray and an arc with the vertex as centre
 - B. draw a perpendicular to the ray
 - C. bisect a right angle
 - D. measure the angle with a protractor
4. When an angle of 135° is bisected, each of the two angles formed is
 - A. 45°
 - B. 70°
 - C. 65°
 - D. $67\frac{1}{2}^\circ$
6. In constructing an angle of 120° with a pair of compasses, after drawing an arc with the vertex as centre, the next step is to
 - A. draw a line through the vertex
 - B. measure the angle with a protractor
 - C. draw a perpendicular to the ray
 - D. use the same radius to step off two equal arcs along the first arc
8. An angle of 135° can be constructed as the sum of
 - A. 100° and 35°
 - B. 90° and 45°
 - C. 80° and 55°
 - D. 120° and 25°
10. An angle of 75° can be constructed by bisecting the angle between rays making
 - A. 60° and 90°
 - B. 30° and 90°
 - C. 45° and 90°
 - D. 60° and 120°

D. one side

- 13.** To construct a triangle when all three sides are given, the construction is done by
- using a protractor at each end of the base
 - drawing arcs from both ends of the base with radii equal to the other two sides
 - bisecting the base
 - drawing perpendiculars at the ends of the base
- 15.** To construct an isosceles triangle with base 6 cm and base angles 50° each, the step after drawing the base is to
- bisect the base
 - draw an arc of radius 6 cm
 - construct 50° at each end of the base
 - construct 80° at one end of the base
- 17.** A triangle with angles 35° , 55° and 90° is classified as
- acute-angled
 - right-angled
 - obtuse-angled
 - equilateral
- 19.** In constructing a triangle given two sides and the included angle, the angle is constructed
- opposite the longer side
 - after drawing all three sides
 - at any convenient position
 - between the two given sides
- 21.** Which of the following is enough to construct a unique triangle?
- Three angles only
 - One side and one angle
 - Three sides
 - Two angles only
- 23.** For three line segments to form a triangle, the sum of the lengths of any two of them must be
- equal to the third
 - less than the third
 - twice the third
 - greater than the third
- 12.** An equilateral triangle is constructed with side 6.2 cm. Each angle of the triangle is
- 45°
 - 90°
 - 60°
 - 120°
- 14.** A triangle with sides 6 cm, 8 cm and 7 cm is
- scalene
 - isosceles
 - equilateral
 - right-angled
- 16.** The vertical angle of the isosceles triangle in question 15 is
- 50°
 - 100°
 - 40°
 - 80°
- 18.** Which of the following sets of angles determines an obtuse-angled triangle?
- 25° , 35° , 120°
 - 40° , 60° , 80°
 - 30° , 60° , 90°
 - 50° , 50° , 80°
- 20.** In triangle ABC, $\angle A = 45^\circ$, $AC = 3$ cm and $AB = 5$ cm. After constructing the angle and marking the two sides, the third side BC is obtained by
- measuring it with a ruler only
 - joining B and C with a straight line
 - bisecting angle A
 - drawing a circle with centre A
- 22.** Three angles only (AAA) determine
- infinitely many triangles of different sizes
 - a unique triangle
 - no triangle at all
 - a right-angled triangle only
- 24.** Which of the following sets of lengths cannot form a triangle?
- 6 cm, 7 cm, 8 cm
 - 3 cm, 4 cm, 8 cm
 - 3 cm, 4 cm, 5 cm
 - 5 cm, 5 cm, 6 cm

- 25.** The locus of a point which is always 5 cm from a fixed point O is
- a straight line
 - a circle of radius 10 cm
 - a circle with centre O and radius 5 cm
 - a ray from O
- 27.** The locus of points equidistant from two intersecting straight lines is
- a pair of parallel lines
 - a circle
 - the perpendicular bisector of the lines
 - the pair of angle bisectors of the angles between the lines
- 29.** The centre of a drawn circle can be located by constructing
- the angle bisectors of two chords
 - the perpendicular bisectors of two chords
 - lines through any two points on the circle
 - the perpendicular bisectors of two radii
- 31.** Two points P and Q are 5 cm apart. The number of points which are 3 cm from both P and Q is
- 0
 - 2
 - 1
 - 4
- 33.** The point equidistant from the three sides of a triangle is found at the intersection of the
- angle bisectors
 - perpendicular bisectors
 - medians
 - diagonals
- 35.** A fountain is equidistant from two straight paths which meet at a point. The fountain lies on
- the perpendicular bisector of the paths
 - the angle bisector of the angle between the paths
 - a circle with centre at the meeting point
- 26.** The locus of a point equidistant from two fixed points P and Q is
- the perpendicular bisector of PQ
 - the line PQ
 - a circle with diameter PQ
 - the angle bisector at P
- 28.** The locus of points equidistant from two parallel lines is
- a line perpendicular to both lines
 - a circle between them
 - a line midway between and parallel to both lines
 - two intersecting lines
- 30.** A goat is tied to a post with a rope 4 m long. The region the goat can graze is
- a circle of radius 4 m with the post as centre
 - a circle of diameter 4 m
 - a square of side 4 m
 - a straight line 4 m long
- 32.** The point equidistant from the three vertices of a triangle is found at the intersection of the
- angle bisectors
 - medians
 - altitudes
 - perpendicular bisectors of the sides
- 34.** The locus of points at a fixed distance from a given straight line is
- one parallel line
 - a circle
 - two parallel lines, one on each side of the given line
 - a perpendicular line

D. the longer path

- 36.** A tree is 10 m from a flagpole and also 10 m from a lamp post which is 12 m from the flagpole. Its position is found by
- A. drawing arcs of radius 10 m from both points
 - B. bisecting the line joining the two points
 - C. drawing parallel lines through the two points
 - D. drawing one circle of radius 20

37. Kofi walks so that he is always 6 m from a long straight wall. His path is
- perpendicular to the wall
 - a circle of radius 6 m
 - parallel to the wall and 6 m from it
 - along the wall
39. A triangle has sides 5 cm, 5 cm and 6 cm. It is best described as
- scalene
 - isosceles
 - equilateral
 - right-angled
38. An angle of 150° can be constructed as the sum of
- 120° and 45°
 - 60° and 45°
 - 100° and 50°
 - 90° and 60°
40. A construction combines the locus of points 5 cm from a fixed point O with the locus of points equidistant from two fixed points A and B. The required points are
- where the circle with centre O meets the perpendicular bisector of AB
 - a single circle
 - the angle bisectors of A and B
 - two parallel lines

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Construction of angles.

Question 2

- (a) More angle constructions.

Question 3

- (a) Construction of triangles.

Question 4

- (a) More triangle constructions.

Question 5

- (a) Loci.

Question 6

- (a) Integrated loci.

Chapter 3: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The ratio of the circumference of any circle to its diameter is
 - A. π
 - B. 2π
 - C. r
 - D. 22
2. The circumference of a circle of radius 7 cm (take $\pi = 22/7$) is
 - A. 154 cm
 - B. 44 cm
 - C. 22 cm
 - D. 308 cm
3. The area of a circle of radius 14 cm (take $\pi = 22/7$) is
 - A. 88 cm^2
 - B. 154 cm^2
 - C. 616 cm^2
 - D. 1232 cm^2
4. The area of a semi-circle of radius 7 cm (take $\pi = 22/7$) is
 - A. 77 cm^2
 - B. 154 cm^2
 - C. 38.5 cm^2
 - D. 44 cm^2
5. Two circles have a common centre; the smaller has radius 7 cm and the bigger has radius 14 cm. The area of the region between them (take $\pi = 22/7$) is
 - A. 154 cm^2
 - B. 308 cm^2
 - C. 462 cm^2
 - D. 616 cm^2
6. The formula for the area of a circle is deduced by cutting the circle into many sectors and arranging them to form a
 - A. rectangle
 - B. triangle
 - C. trapezium
 - D. parallelogram
7. In the sector method of deducing the area of a circle, the rectangle formed has length and width approximately
 - A. $2\pi r$ and r
 - B. πr and r
 - C. πr and $2r$
 - D. r^2 and r
8. The circumference of a circle of diameter 10 cm (take $\pi = 3.14$) is
 - A. 62.8 cm
 - B. 15.7 cm
 - C. 78.5 cm
 - D. 31.4 cm
9. The radius of a circle whose circumference is 88 cm (take $\pi = 22/7$) is
 - A. 28 cm
 - B. 7 cm
 - C. 14 cm
 - D. 44 cm
10. The area of a circle whose circumference is 44 cm (take $\pi = 22/7$) is
 - A. 44 cm^2
 - B. 154 cm^2
 - C. 616 cm^2
 - D. 88 cm^2

- 11.** A circular garden of radius 7 m is to be fenced at GH¢2.00 per metre (take $\pi = 22/7$). The cost of the fencing is
- A. GH¢44.00
 - B. GH¢154.00
 - C. GH¢88.00
 - D. GH¢308.00
- 13.** In a right-angled triangle with hypotenuse c and the other two sides a and b ,
- A. $a + b = c$
 - B. $a^2 + b^2 = c^2$
 - C. $a^2 - b^2 = c^2$
 - D. $2a + 2b = 2c$
- 15.** A right-angled triangle has hypotenuse 13 cm and one side 5 cm. The third side is
- A. 18 cm
 - B. 8 cm
 - C. 144 cm
 - D. 12 cm
- 17.** A ladder leans against a vertical wall of height 8 m. The foot of the ladder is 6 m from the wall. The length of the ladder is
- A. 10 m
 - B. 14 m
 - C. 48 m
 - D. 100 m
- 19.** An isosceles triangle has equal sides 10 cm long and a base 12 cm long. The altitude of the triangle is
- A. 6 cm
 - B. 8 cm
 - C. $\sqrt{136}$ cm
 - D. 16 cm
- 21.** The area of the equilateral triangle in question 20, correct to one decimal place, is
- A. 72.0 cm²
 - B. 62.4 cm²
 - C. 124.8 cm²
 - D. 36.0 cm²
- 23.** Yeboah hangs a picture frame of width 15 cm on the wall with a string passing over a nail. The distance from the nail to each edge of the frame is 10 cm. The length of the string is
- 12.** The area of a circle is 154 cm² (take $\pi = 22/7$). Its radius is
- A. 7 cm
 - B. 14 cm
 - C. 49 cm
 - D. 22 cm
- 14.** The two shorter sides of a right-angled triangle are 3 cm and 4 cm. The hypotenuse is
- A. 7 cm
 - B. 12 cm
 - C. 5 cm
 - D. 25 cm
- 16.** A ladder leans against a vertical wall of height 12 m. The foot of the ladder is 5 m from the wall. The length of the ladder is
- A. 17 m
 - B. 13 m
 - C. 7 m
 - D. $\sqrt{119}$ m
- 18.** A boat travels 2 m South and then 9 m East. Its distance from the starting point, correct to one decimal place, is
- A. 11.0 m
 - B. 8.1 m
 - C. 7.0 m
 - D. 9.2 m
- 20.** The side of an equilateral triangle is 12 cm. The height of the triangle, correct to one decimal place, is
- A. 6.0 cm
 - B. 12.0 cm
 - C. 36.0 cm
 - D. 10.4 cm
- 22.** The perimeter of the equilateral triangle in question 20 is
- A. 24 cm
 - B. 48 cm
 - C. 36 cm
 - D. 144 cm

- A. 25 cm
- B. 10 cm
- C. 15 cm
- D. 20 cm

25. Which of the following is a Pythagorean triple?

- A. 5, 12, 14
- B. 6, 7, 10
- C. 8, 15, 17
- D. 9, 12, 16

24. The vertical distance from the nail to the top edge of the frame in question 23, correct to one decimal place, is

- A. 6.6 cm
- B. 7.5 cm
- C. 10.0 cm
- D. 2.5 cm

- 27.** The diagonal of a rectangle of length 12 cm and width 5 cm is
A. 17 cm
B. 60 cm
C. 7 cm
D. 13 cm
- 29.** In a right-angled triangle, $\sin \theta =$
A. adjacent $\div$ hypotenuse
B. opposite $\div$ adjacent
C. opposite $\div$ hypotenuse
D. hypotenuse $\div$ opposite
- 31.** A right-angled triangle has sides 3, 4 and 5, with the hypotenuse 5. If θ is the angle opposite the side 3, then $\sin \theta =$
A. $4/5$
B. $3/4$
C. $3/5$
D. $5/3$
- 33.** For the triangle in question 31, $\tan \theta =$
A. $3/4$
B. $4/3$
C. $3/5$
D. $4/5$
- 35.** A hunter on top of a tower of height 18 m sees a fire at an angle of depression of 30° . The distance from the hunter to the fire is
A. 9 m
B. 18 m
C. 36 m
D. 54 m
- 37.** A circle of radius 7 cm rolls once along a straight line. The distance it travels (take $\pi = 22/7$) is
A. 154 cm
B. 44 cm
C. 22 cm
D. 7 cm
- 39.** The area of a circle is 616 cm^2 (take $\pi = 22/7$). Its circumference is
A. 44 cm
B. 154 cm
C. 88 cm
D. 28 cm
- 26.** A right-angled triangle has sides 6 cm and 8 cm. The area of the square on its hypotenuse is
A. 48 cm^2
B. 100 cm^2
C. 14 cm^2
D. 96 cm^2
- 28.** The hypotenuse of a right-angled triangle is 25 cm and one side is 7 cm. The third side is
A. 24 cm
B. 18 cm
C. 32 cm
D. 576 cm
- 30.** In a right-angled triangle, $\tan \theta =$
A. adjacent $\div$ hypotenuse
B. opposite $\div$ adjacent
C. hypotenuse $\div$ opposite
D. opposite $\div$ hypotenuse
- 32.** For the triangle in question 31, $\cos \theta =$
A. $3/5$
B. $5/4$
C. $3/4$
D. $4/5$
- 34.** The angle which the line of sight makes with the horizontal when an observer looks up at an object is the
A. angle of depression
B. angle of elevation
C. right angle
D. angle of contact
- 36.** A right-angled triangle has hypotenuse 20 cm. If $\sin \theta = 3/5$, the side opposite θ is
A. 12 cm
B. 16 cm
C. 15 cm
D. 9 cm
- 38.** A ladder 10 m long reaches 8 m up a vertical wall. The distance of its foot from the wall is
A. 2 m
B. 18 m
C. 4 m
D. 6 m

- 40.** Kofi walks 8 m North from a flagpole and then 15 m East. His distance from the flagpole is
- A. 17 m
 - B. 23 m
 - C. 7 m
 - D. 120 m

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The circle.

Question 2

- (a) The Pythagoras theorem.

Question 3

- (a) Pythagoras in real life.

Question 4

- (a) Areas of triangles using Pythagoras.

Question 5

- (a) Trigonometric ratios.

Question 6

- (a) Integrated problems.

Chapter 4: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A quantity which has both magnitude and direction is called
 - A. a scalar
 - B. a magnitude
 - C. a vector
 - D. a component
2. The vector from point A to point B is written as
 - A. vector AB
 - B. vector BA
 - C. scalar AB
 - D. magnitude AB
3. A vector with zero magnitude is called the
 - A. unit vector
 - B. equal vector
 - C. parallel vector
 - D. zero vector
4. If $p = (2, -1)$, then $-p =$
 - A. $(2, 1)$
 - B. $(-2, 1)$
 - C. $(-2, -1)$
 - D. $(1, 2)$
5. The vector $(3, 4)$ represents a movement of
 - A. 3 units to the right and 4 units up
 - B. 4 units to the right and 3 units up
 - C. 3 units to the left and 4 units up
 - D. 3 units up and 4 units to the right
6. The vector $(-2, 5)$ represents a movement of
 - A. 2 units to the right and 5 units up
 - B. 2 units to the left and 5 units down
 - C. 5 units to the left and 2 units up
 - D. 2 units to the left and 5 units up
7. If $a = (2, 3)$ and $b = (4, 1)$, then $a + b =$
 - A. $(8, 3)$
 - B. $(6, 3)$
 - C. $(6, 4)$
 - D. $(2, 12)$
8. If $a = (5, -2)$ and $b = (-3, 4)$, then $a + b =$
 - A. $(2, -6)$
 - B. $(2, 2)$
 - C. $(8, 2)$
 - D. $(-15, -8)$
9. If $a = (-1, -4)$ and $b = (-2, 3)$, then $a + b =$
 - A. $(-3, -1)$
 - B. $(1, -7)$
 - C. $(-3, 1)$
 - D. $(3, 1)$
10. If $AB = (3, 2)$ and $BC = (1, 5)$, then $AC =$
 - A. $(2, -3)$
 - B. $(3, 10)$
 - C. $(7, 4)$
 - D. $(4, 7)$
11. If $PQ = (-2, 6)$ and $QR = (5, -1)$, then $PR =$
 - A. $(7, -7)$
 - B. $(3, 5)$
 - C. $(-3, -5)$

D. $(10, -6)$

13. If $XY = (4, -1)$ and $XZ = (6, 3)$, then
 $YZ =$

- A. $(2, 4)$
- B. $(10, 2)$
- C. $(-2, -4)$
- D. $(2, -4)$

12. If $a + (2, -3) = (5, 4)$, then $a =$

- A. $(7, 1)$
- B. $(3, 1)$
- C. $(3, 7)$
- D. $(-3, -7)$

- 15.** If $a = (4, -3)$, then $2a =$
A. $(6, -1)$
B. $(8, -6)$
C. $(8, 6)$
D. $(2, -5)$
- 17.** If $q = (3, 4)$, then $-2q =$
A. $(1, 2)$
B. $(6, 8)$
C. $(-6, -8)$
D. $(-6, 8)$
- 19.** If $p = (2, -1)$ and $r = (-2, 3)$, then $r - 3p =$
A. $(4, 0)$
B. $(-8, -6)$
C. $(4, 6)$
D. $(-8, 6)$
- 21.** If $a = (-3, 2)$, then $-a + 2a =$
A. $(-6, 4)$
B. $(0, 0)$
C. $(-9, 6)$
D. $(-3, 2)$
- 23.** If $k(x, y) = (12, -8)$ and $k = 4$, then $(x, y) =$
A. $(3, -2)$
B. $(16, -4)$
C. $(48, -32)$
D. $(-3, 2)$
- 25.** If $AB = (4, 2)$, then $BA =$
A. $(4, 2)$
B. $(2, 4)$
C. $(4, -2)$
D. $(-4, -2)$
- 27.** Two vectors are equal if and only if they have the same
A. starting point
B. magnitude and direction
C. magnitude only
D. direction only
- 29.** If $(2x, 3) = (8, y)$, then $x + y =$
A. 11
B. 5
C. 6
D. 7
- 14.** If $a = (1, 2)$, $b = (3, -1)$ and $c = (-2, 4)$, then $a + b + c =$
A. $(6, 5)$
B. $(2, -3)$
C. $(4, 3)$
D. $(2, 5)$
- 16.** If $p = (2, -1)$, then $3p =$
A. $(6, -3)$
B. $(5, 2)$
C. $(6, 3)$
D. $(-6, -3)$
- 18.** If $p = (2, -1)$ and $q = (3, 4)$, then $3q - 2p =$
A. $(13, 10)$
B. $(5, 14)$
C. $(5, 10)$
D. $(-1, -14)$
- 20.** If $p = (2, -1)$, $q = (3, 4)$ and $r = (-2, 3)$, then $q - p + 2r =$
A. $(-3, 11)$
B. $(1, 9)$
C. $(-3, 5)$
D. $(7, 1)$
- 22.** If $a = (4, -6)$, then $\frac{1}{2}a =$
A. $(4, -3)$
B. $(2, -3)$
C. $(8, -12)$
D. $(2, 3)$
- 24.** If $a = (2, 1)$ and $b = (1, -3)$, then $2a - 3b =$
A. $(1, -7)$
B. $(7, 11)$
C. $(1, 11)$
D. $(-1, -11)$
- 26.** If $a = (-2, 5)$ and $b = (3, -1)$, then $b - a =$
A. $(1, 4)$
B. $(-5, 6)$
C. $(5, -6)$
D. $(-1, -4)$
- 28.** If $(x, 4) = (7, y)$, then
A. $x = 7, y = 4$
B. $x = 4, y = 7$
C. $x = 7, y = -4$
D. $x = -7, y = 4$

- 31.** If $2(x, y) = (6, -10)$, then $(x, y) =$
- A. $(3, -5)$
 - B. $(8, -8)$
 - C. $(12, -20)$
 - D. $(-3, 5)$
- 33.** A man walks 3 km East and then 4 km North. The magnitude of his resultant displacement is
- A. 7 km
 - B. 5 km
 - C. 12 km
 - D. 25 km
- 35.** A girl's total displacement after two walks is $(6, -2)$. If the first walk was $(2, 3)$, the second walk was
- A. $(4, -5)$
 - B. $(8, 1)$
 - C. $(-4, 5)$
 - D. $(4, 5)$
- 37.** If $AB = (a, b)$ and $BC = (c, d)$, then $AB - BC =$
- A. $(a + c, b + d)$
 - B. $(c - a, d - b)$
 - C. $(a - c, d - b)$
 - D. $(a - c, b - d)$
- 39.** If $k(3, -2) = (-9, 6)$, the value of k is
- A. -3
 - B. 3
 - C. -6
 - D. $-1/3$
- 30.** If $(x + 2, y - 1) = (5, 3)$, then
- A. $x = 7, y = 2$
 - B. $x = 3, y = 2$
 - C. $x = 3, y = 4$
 - D. $x = 2, y = 3$
- 32.** If $a = (3, -2)$ and $b = (x + 1, y)$, and $a = b$, then
- A. $x = 3, y = -2$
 - B. $x = 4, y = -2$
 - C. $x = 2, y = 2$
 - D. $x = 2, y = -2$
- 34.** A boat sails $(2, 5)$ km and then $(-1, 3)$ km. Its resultant displacement is
- A. $(3, 2)$
 - B. $(1, 2)$
 - C. $(1, 8)$
 - D. $(-2, 15)$
- 36.** If $AB = (a, b)$ and $BC = (c, d)$, then $AC =$
- A. $(a - c, b - d)$
 - B. $(a + c, b + d)$
 - C. (ac, bd)
 - D. $(a + c, b - d)$
- 38.** The position of B relative to A is given by the vector $(5, -3)$. The position of A relative to B is
- A. $(5, 3)$
 - B. $(3, -5)$
 - C. $(-5, 3)$
 - D. $(-5, -3)$
- 40.** If $a = (2, 4)$ and $b = (1, 2)$, then
- A. $a = b$
 - B. $a = 2b$
 - C. $b = 2a$
 - D. $a + b = 0$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Vector concepts.

Question 2

(a) Addition of vectors.

Question 3

(a) Subtraction and scalar multiplication.

Question 4

(a) Combined operations.

Question 5

(a) Vector equality.

Question 6

(a) Applications.

Chapter 5: Ch14

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following is an example of rotation in everyday life?
 - A. a book sliding along a table
 - B. a stone falling from a height
 - C. a child walking along a straight path
 - D. the blades of a turning fan
2. A rotation is fully described by its
 - A. centre and distance only
 - B. centre, angle and direction
 - C. angle and length only
 - D. direction and distance only
3. The fixed point about which a shape turns is called the
 - A. centre of rotation
 - B. angle of rotation
 - C. line of symmetry
 - D. point of origin
4. The hands of a clock move
 - A. anticlockwise
 - B. in both directions
 - C. clockwise
 - D. along a straight line
5. A quarter turn corresponds to an angle of
 - A. 45°
 - B. 180°
 - C. 360°
 - D. 90°
6. A half turn corresponds to an angle of
 - A. 90°
 - B. 180°
 - C. 270°
 - D. 360°
7. A complete turn corresponds to an angle of
 - A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
8. A steering wheel of a car and the blades of a ceiling fan both demonstrate
 - A. rotation
 - B. translation
 - C. reflection
 - D. enlargement
9. The image of the point (2, 3) under a rotation of 90° anticlockwise about the origin is
 - A. (3, 2)
 - B. (-2, -3)
 - C. (3, -2)
 - D. (-3, 2)
10. The image of the point (4, -1) under a rotation of 90° anticlockwise about the origin is
 - A. (-1, -4)
 - B. (1, 4)
 - C. (-4, -1)
 - D. (1, -4)
11. The image of the point (5, 2) under a rotation of 90° clockwise about the origin is
 - A. (2, -5)
 - B. (-2, 5)
 - C. (-5, -2)
 - D. (5, -2)
12. The image of the point (-3, 4) under a rotation of 180° about the origin is
 - A. (-3, -4)
 - B. (4, 3)
 - C. (-4, -3)
 - D. (3, -4)

- 13.** The image of the point $(-2, -5)$ under a rotation of 180° about the origin is
- A. $(-5, -2)$
 - B. $(5, 2)$
 - C. $(2, 5)$
 - D. $(-2, 5)$
- 15.** The image of the point $(-1, -4)$ under a rotation of 90° anticlockwise about the origin is
- A. $(4, -1)$
 - B. $(-4, 1)$
 - C. $(1, 4)$
 - D. $(-4, -1)$
- 17.** Under a rotation of 90° anticlockwise about the origin, the rule is $(x, y) \rightarrow$
- A. $(y, -x)$
 - B. $(-y, x)$
 - C. $(-x, -y)$
 - D. (y, x)
- 19.** The point $(3, 1)$ is rotated 90° clockwise about the origin. Its image lies in the
- A. fourth quadrant
 - B. first quadrant
 - C. second quadrant
 - D. third quadrant
- 21.** The vertices of triangle ABC are A(1, 0), B(3, 0) and C(1, 2). Under a rotation of 90° anticlockwise about the origin, the image of C is
- A. $(2, 1)$
 - B. $(-1, -2)$
 - C. $(2, -1)$
 - D. $(-2, 1)$
- 23.** Under a rotation about the origin, the point $(2, 7)$ maps onto $(-7, 2)$. The rotation is
- A. 90° clockwise
 - B. 90° anticlockwise
 - C. 180°
 - D. 270° anticlockwise
- 25.** The angle of rotation is found by joining the centre of rotation to a point and its image, then measuring the angle between
- A. the point and the origin
- 14.** The image of the point $(6, -3)$ under a rotation of 270° anticlockwise about the origin is
- A. $(3, 6)$
 - B. $(-3, -6)$
 - C. $(-6, 3)$
 - D. $(6, 3)$
- 16.** A point P is rotated through 360° about the origin. Its image is
- A. the reflection of P
 - B. the point opposite P through the origin
 - C. P itself
 - D. the origin
- 18.** Under a rotation of 180° about the origin, the rule is $(x, y) \rightarrow$
- A. (y, x)
 - B. $(-y, x)$
 - C. $(y, -x)$
 - D. $(-x, -y)$
- 20.** The point $(-2, 5)$ is rotated 180° about the origin. Its image lies in the
- A. first quadrant
 - B. second quadrant
 - C. fourth quadrant
 - D. third quadrant
- 22.** The image of the point (a, b) under a rotation of 90° clockwise about the origin is
- A. $(b, -a)$
 - B. $(-b, a)$
 - C. $(-a, -b)$
 - D. (b, a)
- 24.** Under a rotation about the origin, the point $(3, -2)$ maps onto $(-3, 2)$. The rotation is
- A. 90° clockwise
 - B. 90° anticlockwise
 - C. 180°
 - D. 360°

- B. the image and the x-axis
- C. the object and the y-axis
- D. the two joining lines

27. The minute hand of a clock moves from 12 to 3. The angle and direction of rotation are

- A. 90° clockwise
- B. 90° anticlockwise
- C. 180° clockwise
- D. 270° clockwise

26. A wheel turns through three quarter-turns. The total angle of rotation is

- A. 180°
- B. 270°
- C. 90°
- D. 360°

- 29.** A shape is rotated so that a point and its image subtend a right angle at the centre of rotation. The angle of rotation is
- A. 45°
 - B. 180°
 - C. 90°
 - D. 360°
- 31.** The point (2, 0) is rotated about the origin onto (0, 2). The anticlockwise angle of rotation is
- A. 90°
 - B. 180°
 - C. 270°
 - D. 45°
- 33.** After a rotation, the image of a shape is always
- A. larger than the object
 - B. congruent to the object
 - C. smaller than the object
 - D. a different shape
- 35.** Which of the following is preserved under a rotation?
- A. the lengths of sides and the sizes of angles
 - B. the coordinates of the vertices
 - C. the position of the shape
 - D. the direction the shape faces
- 37.** Triangle PQR is rotated onto triangle P'Q'R'. If PQ = 5 cm, then P'Q' =
- A. 2.5 cm
 - B. 10 cm
 - C. 25 cm
 - D. 5 cm
- 39.** Which of the following transformations always maps a shape onto a congruent image?
- A. enlargement
 - B. rotation
 - C. stretching
 - D. shrinking
- 28.** The minute hand of a clock moves from 12 to 8. The clockwise angle through which it turns is
- A. 120°
 - B. 160°
 - C. 200°
 - D. 240°
- 30.** A fan turns through 540° . The number of complete turns made is
- A. $1\frac{1}{4}$
 - B. $1\frac{1}{2}$
 - C. 2
 - D. $\frac{3}{4}$
- 32.** The point (3, 0) is rotated about the origin onto (0, -3). The anticlockwise angle of rotation is
- A. 90°
 - B. 180°
 - C. 270°
 - D. 360°
- 34.** Under a rotation, corresponding sides of the object and its image are
- A. parallel
 - B. perpendicular
 - C. doubled
 - D. equal in length
- 36.** Two shapes are congruent if they
- A. have the same area only
 - B. are enlargements of each other
 - C. have exactly the same shape and size
 - D. have the same perimeter only
- 38.** In question 37, if $\angle P = 40^\circ$, then $\angle P' =$
- A. 40°
 - B. 50°
 - C. 80°
 - D. 140°
- 40.** Under a rotation, which of the following changes?
- A. the lengths of the sides
 - B. the sizes of the angles
 - C. the position of the shape
 - D. the area of the shape

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Rotation concepts.

Question 2

(a) Rotation rules about the origin.

Question 3

(a) Rotating a shape.

Question 4

(a) Angle of rotation.

Question 5

(a) Congruence.

Question 6

(a) Integrated problems.

Chapter 6: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Data which can be counted and takes whole-number values only is called
 - A. discrete data
 - B. continuous data
 - C. categorical data
 - D. grouped data
2. Which of the following is an example of continuous data?
 - A. the number of children in a family
 - B. the number of learners in a class
 - C. the type of movie watched
 - D. the weights of babies in a crèche
3. Which of the following is an example of discrete data?
 - A. the height of a student
 - B. the number of cars sold in a year
 - C. the time taken to run a race
 - D. the amount of rainfall in a month
4. Which of the following is categorical (qualitative) data?
 - A. the number of learners in a B8 class
 - B. scores in a class test
 - C. the marital status of a person
 - D. the heights of students
5. Data collected directly, before it is organised or processed, is called
 - A. raw data
 - B. grouped data
 - C. secondary data
 - D. information
6. Data arranged into classes or intervals is called
 - A. raw data
 - B. discrete data
 - C. grouped data
 - D. categorical data
7. Data which can take any value within a given range is called
 - A. discrete data
 - B. continuous data
 - C. categorical data
 - D. qualitative data
8. 'Sex' (male or female) is an example of
 - A. categorical data
 - B. continuous data
 - C. discrete data
 - D. grouped data
9. The scores of 11 learners in a class test are 25, 30, 35, 40, 45, 26, 29, 50, 45, 37 and 47. These scores are best described as
 - A. categorical data
 - B. grouped data
 - C. continuous data
 - D. ungrouped numerical (discrete) data
10. Boxers' weight classes and age groups are examples of non-numeric data which can be placed on
 - A. a pie chart only
 - B. a number line
 - C. an ordinal scale
 - D. a tally chart only
11. The best method of collecting data on the favourite colours of learners in a school is
 - A. an experiment

- B. a questionnaire
- C. weighing
- D. measuring

- 12.** To find out whether eating twice a person's normal number of cream crackers increases productivity, the best method of data collection is

- 13.** Ama wants to know how people in her community obtain drinking water. The most appropriate method of data collection is
- A. measuring their heights
 - B. weighing them
 - C. using a stem-and-leaf plot
 - D. an interview (questionnaire)
- 15.** A printed form designed to collect information from respondents is called
- A. a bar chart
 - B. a questionnaire
 - C. a tally chart
 - D. a waffle chart
- 17.** A tally chart is used to
- A. measure heights
 - B. calculate averages
 - C. record frequencies using strokes
 - D. draw circles
- 19.** The lengths, in millimetres, of 40 leaves were measured to the nearest mm and grouped into classes 25–29, 30–34, 35–39, 40–44, 45–49, 50–54 and 55–59. If the class 25–29 contains the leaves 25 mm and 28 mm only, its frequency is
- A. 2
 - B. 5
 - C. 25
 - D. 28
- 21.** Using the leaf-length data, the modal class is
- A. 40–44
 - B. 25–29
 - C. 45–49
 - D. 55–59
- 23.** Using the leaf-length data, the total number of leaves measured is
- A. 36
 - B. 38
 - C. 39
 - D. 40
- A. an experiment
B. observation only
C. a pie chart
D. a pictogram
- 14.** Data collected by watching and recording events without changing anything is collected by
- A. experiment
 - B. questionnaire
 - C. observation
 - D. simulation
- 16.** Which of the following questions is suitable for a quantitative study?
- A. What is your favourite song?
 - B. Which movie do you enjoy most?
 - C. What is your religion?
 - D. How many hours do you study each day?
- 18.** The tally group of four vertical strokes crossed by a diagonal stroke represents
- A. 4
 - B. 5
 - C. 9
 - D. 10
- 20.** Using the leaf-length data, the frequency of the class 40–44 is
- A. 7
 - B. 8
 - C. 4
 - D. 9
- 22.** Using the leaf-length data, the number of leaves with lengths of at least 50 mm is
- A. 7
 - B. 10
 - C. 3
 - D. 13
- 24.** The class width of the classes 25–29, 30–34 and 35–39 is
- A. 4
 - B. 29
 - C. 5
 - D. 25

- 25.** In a stem-and-leaf plot of test scores, the entry '4 | 7' represents the score
A. 47
B. 74
C. 4.7
D. 407
- 27.** In a bar chart, the height of each bar represents the
A. frequency
B. class width
C. tally
D. percentage only
- 29.** The total angle at the centre of a complete pie chart is
A. 90°
B. 180°
C. 270°
D. 360°
- 31.** From the cleaner's data in question 30, the sector angle for food in a pie chart is
A. 72°
B. 108°
C. 120°
D. 99°
- 33.** From the cleaner's data in question 30, the sector angle for clothing in a pie chart is
A. 40°
B. 50°
C. 36°
D. 72°
- 35.** In a pictogram, one symbol represents 5 learners. The number of symbols needed to represent 35 learners is
A. 5
B. 7
C. 35
D. 175
- 37.** A waffle chart shows that the average score of B7 learners in a mathematics test is 64%. The number of shaded cells is
A. 64
B. 36
C. 6.4
D. 100
- 26.** In a stem-and-leaf plot, if no score falls in the 30s, the row for stem 3 should be
A. written with a 0 against it
B. filled with 3s
C. left blank
D. removed from the plot
- 28.** A bar chart is most suitable for displaying data which are
A. only continuous
B. categorical or discrete
C. only grouped
D. only percentages
- 30.** A cleaner of a small office spent GH¢120.00 on food, GH¢80.00 on rent, GH¢40.00 on clothing, GH¢110.00 on transport and saved GH¢50.00. The total amount spent and saved is
A. GH¢350.00
B. GH¢360.00
C. GH¢400.00
D. GH¢440.00
- 32.** From the cleaner's data in question 30, the sector angle for transport in a pie chart is
A. 45°
B. 36°
C. 110°
D. 99°
- 34.** From the cleaner's data in question 30, the fraction of the total amount which was spent on rent is
A. $\frac{1}{5}$
B. $\frac{1}{4}$
C. $\frac{2}{5}$
D. $\frac{1}{3}$
- 36.** In a waffle chart of 100 cells, each cell represents
A. 10%
B. 5%
C. 100%
D. 1%
- 38.** In a pictogram, one full symbol represents 10 mangoes. The number of mangoes represented by $3\frac{1}{2}$ symbols is
A. 3.5

- B. 30
- C. 35
- D. 70

39. Which of the following displays is best for showing how a whole quantity is shared among its parts?

- A. a stem-and-leaf plot
- B. a pie chart
- C. a tally chart
- D. a scatter diagram

40. The data 12, 15, 15, 23, 27 and 31 are displayed in a stem-and-leaf plot. The number of leaves against stem 1 is

- A. 2
- B. 6
- C. 1
- D. 3

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Types of data.

Question 2

(a) Methods of data collection.

Question 3

(a) Frequency tables.

Question 4

(a) Bar charts and pie charts.

Question 5

(a) Stem-and-leaf plots, waffle charts and pictograms.

Question 6

(a) Integrated problems.

Chapter 7: Ch16

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mean of 4, 7, 9 and 12 is
 - A. 32
 - B. 8
 - C. 7
 - D. 10
2. The mean of 5, 5, 8 and 10 is
 - A. 28
 - B. 6
 - C. 7
 - D. 8
3. The sum of six numbers is 48. Their mean is
 - A. 8
 - B. 288
 - C. 6
 - D. 42
4. The mean of three numbers is 10. Two of the numbers are 8 and 13. The third number is
 - A. 10
 - B. 31
 - C. 11
 - D. 9
5. The mean of x , 5, 7 and 10 is 7. The value of x is
 - A. 5
 - B. 6
 - C. 7
 - D. 28
6. Ama's scores in five tests are 12, 15, 14, 16 and 13. Her mean score is
 - A. 14
 - B. 70
 - C. 13
 - D. 15
7. The mean of 2.5, 3.5 and 6 is
 - A. 3
 - B. 12
 - C. 3.5
 - D. 4
8. Kojo's taxi made 8, 6, 10, 10, 9, 10 and 3 trips from Monday to Sunday. The mean number of trips per day is
 - A. 56
 - B. 10
 - C. 8
 - D. 9
9. The mean of a set of ten values is 6. If one value of 6 is removed, the mean of the remaining values is
 - A. 5.4
 - B. 6
 - C. 6.6
 - D. cannot be determined
10. The mean height of four pupils is 140 cm. When a fifth pupil joins the group, the mean becomes 142 cm. The height of the fifth pupil is
 - A. 150 cm
 - B. 142 cm
 - C. 148 cm
 - D. 154 cm
11. The mode of 2, 3, 3, 5, 7, 3 and 8 is
 - A. 5
 - B. 8
 - C. 2
 - D. 3
12. The median of 4, 9, 1, 7 and 5 is
 - A. 4
 - B. 5
 - C. 7
 - D. 9

- 13.** The median of 2, 4, 6, 8, 10 and 12 is
A. 7
B. 6
C. 8
D. 7.5
- 15.** For the data set 3, 5, 5, 7, 9, which of the following is true?
A. the mean equals the median
B. the median is less than the mode
C. the mean equals the mode
D. the median equals the mode
- 17.** For the house-area data in question 16, the median area is
A. 30 m^2
B. 26 m^2
C. 48 m^2
D. 24 m^2
- 19.** For the house-area data in question 16, the mean, median and mode are all equal because
A. the data contain an outlier
B. the data are balanced around 30
C. all seven values are equal
D. 48 appears only once
- 21.** The range of 4, 9, 2, 11 and 6 is
A. 9
B. 32
C. 11
D. 2
- 23.** The greatest value in a data set is 25 and the range is 18. The smallest value is
A. 43
B. 18
C. 7
D. 25
- 25.** The measure which uses only the greatest and the smallest values of a data set is the
A. range
B. mean
C. median
D. mode
- 27.** The measure of central tendency most affected by an extreme value (outlier) is the
- 14.** The mode of 1, 2, 2, 3, 4, 4 and 5 is
A. 2 only
B. 4 only
C. 2 and 4
D. there is no mode
- 16.** The areas of the sitting-room floors of seven houses are 22, 24, 26, 30, 48, 30 and 30 m^2 . The mean area is
A. 26 m^2
B. 210 m^2
C. 30 m^2
D. 48 m^2
- 18.** For the house-area data in question 16, the mode is
A. 22 m^2
B. 26 m^2
C. 48 m^2
D. 30 m^2
- 20.** In a frequency table, the data value 2 has the greatest frequency. The mode of the data is
A. the greatest frequency
B. the sum of the frequencies
C. 2
D. 11
- 22.** The range of Kojo's taxi trips in question 8 is
A. 10
B. 7
C. 3
D. 56
- 24.** The range of 5, 5, 5 and 5 is
A. 5
B. 20
C. 1
D. 0
- 26.** The range of the house-area data in question 16 is
A. 30 m^2
B. 22 m^2
C. 48 m^2
D. 26 m^2

- A. mode
- B. median
- C. mean
- D. range

- 28.** The most appropriate measure of central tendency for finding the most popular shoe size in a shop is the
- A. mean

- B. mode
C. median
D. range
- 29.** The most appropriate measure for describing a typical income in a community where a few people earn far more than the rest is the
- A. median
 - B. mean
 - C. mode
 - D. range
- 31.** The median of Kojo's taxi trips in question 8 is
- A. 8
 - B. 9
 - C. 10
 - D. 3
- 33.** The measure which best represents the typical number of trips Kojo makes each day is
- A. the mean, because it uses all the values
 - B. the range
 - C. the mode, because 10 is the greatest value
 - D. the median, because it is not pulled down by the low value of 3
- 35.** A frequency table shows the values 1 to 7 with frequencies 1, 11, 5, 4, 2, 1 and 1. The total number of data values is
- A. 7
 - B. 25
 - C. 11
 - D. 77
- 37.** For the frequency table in question 35, the median of the data is
- A. 2
 - B. 11
 - C. 4
 - D. 3
- 39.** For the frequency table in question 35, the mean, median and mode are different because
- A. the data are symmetric
 - B. the data are skewed, with most
- 30.** The most appropriate measure to use when every value in the data set matters equally is the
- A. mode
 - B. range
 - C. median
 - D. mean
- 32.** The mode of Kojo's taxi trips in question 8 is
- A. 10
 - B. 9
 - C. 8
 - D. 56
- 34.** In a class test most pupils scored around 15 marks, but one pupil scored 0 because she was absent. The teacher reporting the class performance should use
- A. the mean, because it is exact
 - B. the range
 - C. the median or the mode, because the 0 is an extreme value
 - D. any of the three, because they are all equally suitable
- 36.** For the frequency table in question 35, the mean of the data is
- A. 3
 - B. 4
 - C. 3.08
 - D. 77
- 38.** For the frequency table in question 35, the mode of the data is
- A. 2
 - B. 11
 - C. 3
 - D. 1

values small

- C. all the frequencies are equal
- D. no value is repeated

- 40.** The mean, median and mode of a data set are all equal. The data set is most likely
- A. skewed to the left
 - B. skewed to the right
 - C. symmetric
 - D. full of outliers

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) The mean.

Question 2

(a) Median, mode and range.

Question 3

(a) The house-area data.

Question 4

(a) Kojo's taxi trips.

Question 5

(a) A frequency table.

Question 6

(a) Choosing and creating.

Chapter 8: Ch17

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The set of all possible outcomes of a probability experiment is called the
 - A. event
 - B. probability
 - C. trial
 - D. sample space
2. The sample space for tossing a fair coin once is
 - A. {H, T}
 - B. {H}
 - C. {H, H, T}
 - D. {1, 2}
3. The sample space for throwing a fair six-sided die has how many outcomes?
 - A. 2
 - B. 3
 - C. 6
 - D. 12
4. Two coins are tossed once. The sample space is
 - A. {HH, TT}
 - B. {HH, HT, TH, TT}
 - C. {HH, HT, TT}
 - D. {H, T}
5. A bag contains 3 red and 2 green bottle tops. One bottle top is picked at random. The number of possible colour outcomes is
 - A. 1
 - B. 3
 - C. 5
 - D. 2
6. A die is thrown and a coin is tossed. The number of outcomes in the sample space is
 - A. 8
 - B. 12
 - C. 6
 - D. 36
7. Two events are independent if the outcome of one
 - A. does not affect the outcome of the other
 - B. determines the outcome of the other
 - C. is always equal to the outcome of the other
 - D. always happens before the other
8. Ampofo walks to school daily and Serwaa travels to school by bus daily. The event 'Ampofo walks' and the event 'Serwaa travels by bus' are
 - A. dependent events
 - B. the same event
 - C. impossible events
 - D. independent events
9. A bottle top is picked from a bag, replaced, and a second bottle top is picked. The two picks are
 - A. dependent
 - B. impossible
 - C. independent
 - D. certain
10. A coin is tossed and a spinner numbered 1 and 2 is spun. The number of outcomes in the sample space is
 - A. 2
 - B. 4
 - C. 6
 - D. 8

- 11.** A box contains 3 blue pens and 4 pink pens. A pen is taken, its colour noted and replaced, and another pen is taken. The sample space of the two trials is
- A. {BB, BP, PB, PP}
 - B. {BB, PP}
 - C. {B, P}
 - D. {BB, BP, PP}
- 13.** A fair die is thrown. The probability of scoring a 6 is
- A. $\frac{1}{2}$
 - B. $\frac{1}{6}$
 - C. 6
 - D. $\frac{1}{3}$
- 15.** A bag contains 3 red, 2 green and 1 pink bottle top. One bottle top is picked at random. The probability of picking a red bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{6}$
 - D. $\frac{2}{3}$
- 17.** From the bag in question 15, the probability of picking the pink bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{4}$
 - D. $\frac{1}{6}$
- 19.** A box contains 3 blue pens and 4 pink pens. One pen is taken at random. The probability that it is blue is
- A. $\frac{3}{4}$
 - B. $\frac{3}{7}$
 - C. $\frac{4}{7}$
 - D. $\frac{1}{7}$
- 21.** A pen is taken from the box in question 19, replaced, and a second pen is taken. The probability that both pens are blue is
- A. $\frac{3}{7}$
 - B. $\frac{6}{7}$
 - C. $\frac{6}{49}$
 - D. $\frac{9}{49}$
- 23.** From the experiment in question 21, the probability of taking one blue pen
- 12.** A spinner with 3 equal sections is spun twice. The number of possible outcomes is
- A. 3
 - B. 6
 - C. 9
 - D. 12
- 14.** A fair die is thrown. The probability of scoring an even number is
- A. $\frac{1}{6}$
 - B. $\frac{1}{3}$
 - C. $\frac{2}{3}$
 - D. $\frac{1}{2}$
- 16.** From the bag in question 15, the probability of picking a green bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{6}$
 - C. $\frac{1}{3}$
 - D. $\frac{2}{5}$
- 18.** From the bag in question 15, the probability of NOT picking a red bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{6}$
 - D. $\frac{5}{6}$
- 20.** From the box in question 19, the probability that the pen taken is pink is
- A. $\frac{3}{7}$
 - B. $\frac{4}{3}$
 - C. $\frac{4}{7}$
 - D. $\frac{1}{4}$
- 22.** From the experiment in question 21, the probability that both pens are pink is
- A. $\frac{12}{49}$
 - B. $\frac{16}{49}$
 - C. $\frac{4}{7}$
 - D. $\frac{8}{49}$

and one pink pen, in any order, is

- A. $\frac{24}{49}$
- B. $\frac{12}{49}$
- C. $\frac{9}{49}$
- D. $\frac{25}{49}$

25. The probability of an impossible event is

- A. 1
- B. $\frac{1}{2}$
- C. between 0 and 1
- D. 0

24. The probability of an event which is certain to happen is

- A. 0
- B. $\frac{1}{2}$
- C. 1
- D. greater than 1

- 27.** The probability $\frac{1}{4}$ expressed as a decimal is
A. 0.4
B. 0.25
C. 4.0
D. 2.5
- 29.** The probability $\frac{1}{4}$ expressed as a ratio of favourable outcomes to total outcomes is
A. 1 : 3
B. 4 : 1
C. 1 : 4
D. 3 : 4
- 31.** The probability 45% expressed as a fraction in its lowest terms is
A. $\frac{45}{10}$
B. $\frac{9}{20}$
C. $\frac{4}{5}$
D. $\frac{9}{10}$
- 33.** The probability of picking a red bottle top from the bag in question 15, expressed as a percentage, is
A. 50%
B. 30%
C. 3%
D. 500%
- 35.** A spinner with two equal sections, Win and Lose, is spun twice. The probability of winning twice is
A. $\frac{1}{2}$
B. $\frac{1}{4}$
C. $\frac{3}{4}$
D. $\frac{1}{8}$
- 37.** From the spinner in question 35, the probability of exactly one win in the two spins is
A. $\frac{1}{2}$
B. $\frac{1}{4}$
C. $\frac{3}{4}$
D. 1
- 26.** A spinner records a win or a loss each time it is spun. If $P(\text{win}) = \frac{3}{5}$, then $P(\text{loss}) =$
A. $\frac{2}{5}$
B. $\frac{3}{5}$
C. $\frac{5}{3}$
D. $\frac{1}{5}$
- 28.** The probability $\frac{1}{4}$ expressed as a percentage is
A. 4%
B. 14%
C. 40%
D. 25%
- 30.** The probability 0.6 expressed as a fraction in its lowest terms is
A. $\frac{3}{5}$
B. $\frac{6}{10}$
C. $\frac{3}{10}$
D. $\frac{5}{3}$
- 32.** The probability $\frac{3}{8}$ expressed as a decimal is
A. 0.83
B. 3.8
C. 0.375
D. 0.308
- 34.** An event has probability $\frac{2}{7}$. The ratio of favourable outcomes to total outcomes is
A. 7 : 2
B. 2 : 5
C. 5 : 7
D. 2 : 7
- 36.** From the spinner in question 35, the probability of losing twice is
A. $\frac{1}{2}$
B. $\frac{3}{4}$
C. $\frac{1}{4}$
D. 0
- 38.** In a probability tree diagram, the probability of two successive events is found by
A. adding the probabilities on the branches
B. multiplying the probabilities on the branches
C. subtracting the probabilities on the branches

D. dividing the probabilities on the branches

40. A die is thrown at most three times, and the game stops when a 6 is scored. The probability that the game stops at the first throw is

- A. $\frac{1}{3}$
- B. $\frac{1}{2}$
- C. $\frac{1}{6}$
- D. 6

39. In a complete probability tree diagram, the sum of the probabilities of all the final outcomes is

- A. 0
- B. $\frac{1}{2}$
- C. 2
- D. 1

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Sample spaces.

Question 2

(a) The bottle-top experiment.

Question 3

(a) The pen experiment.

Question 4

(a) The spinner experiment.

Question 5

(a) Conversions.

Question 6

(a) Integrated problems.

Chapter 9: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The place value of the digit 6 in the number 2,468,321,507 is
 - A. 6,000,000
 - B. 600,000
 - C. 60,000,000
 - D. 6,000
2. The number 3,000,000,000 written in words is
 - A. three billion
 - B. three million
 - C. thirty billion
 - D. three hundred million
3. Two million, four hundred and eight thousand, three hundred and twenty-one written in figures is
 - A. 2,480,321
 - B. 2,408,231
 - C. 2,048,321
 - D. 2,408,321
4. The value of the digit 5 in the numeral part of the serial number TD1,567,451 is
 - A. 5,000
 - B. 500,000
 - C. 50,000
 - D. 5,000,000
5. How many millions make one billion?
 - A. 1,000
 - B. 100
 - C. 10
 - D. 1,000,000
6. The figure 4,050,006,000 written in words is
 - A. four billion, five million and six thousand
 - B. four billion, fifty million and six thousand
 - C. forty billion, five million and sixty thousand
 - D. four billion, fifty million and six
7. The number of zeros in one billion is
 - A. 6
 - B. 8
 - C. 9
 - D. 12
8. In the number 7,891,234,560, the digit in the hundred millions place is
 - A. 8
 - B. 9
 - C. 7
 - D. 1
9. Counting forwards in 500,000s beginning from 200,000, the 3rd number is
 - A. 700,000
 - B. 1,000,000
 - C. 1,500,000
 - D. 1,200,000
10. Counting backwards in 100,050s beginning from 1,800,000, the 3rd number is
 - A. 1,700,000
 - B. 1,699,950
 - C. 1,599,900
 - D. 1,500,000
11. Counting forwards in 500,000s, the number that comes immediately after 2,500,000 is
 - A. 2,550,000
 - B. 3,000,000
 - C. 3,500,000
 - D. 2,000,000

- 13.** In a skip count, the number after 750,000 is 800,000. The counting is done in
A. 5,000s
B. 100,000s
C. 50,000s
D. 25,000s
- 15.** Which of the following numbers is the greatest?
A. 9,999,999
B. 1,000,000,001
C. 1,000,000,000
D. 999,999,999
- 17.** The numbers 4,050,000,000; 4,005,000,000 and 4,500,000,000 arranged in ascending order are
A. 4,050,000,000; 4,005,000,000; 4,500,000,000
B. 4,500,000,000; 4,050,000,000; 4,005,000,000
C. 4,005,000,000; 4,500,000,000; 4,050,000,000
D. 4,005,000,000; 4,050,000,000; 4,500,000,000
- 19.** Which of the following numbers lies between 2,400,000,000 and 2,500,000,000?
A. 2,390,000,000
B. 2,450,000,000
C. 2,510,000,000
D. 2,550,000,000
- 21.** The standard form of 26 is
A. 2.6×10
B. 2.6×10^2
C. 0.26×10^2
D. 26×10
- 23.** The standard form of 8,650,049 is
A. 8.650049×10^5
B. 8.650049×10^6
C. 86.50049×10^5
D. 8.650049×10^7
- 25.** Written as an ordinary number, 4.2×10^5 is
A. 42,000
B. 4,200,000
C. 4,200
D. 420,000
- 12.** Counting backwards in 10,000s beginning from 1,000,000, the 5th number is
A. 960,000
B. 950,000
C. 970,000
D. 940,000
- 14.** Which symbol makes the statement true? $3,450,000,000 \square 3,405,000,000$
A. $<$
B. $=$
C. $\leq$
D. $>$
- 16.** Which of the following numbers is the smallest?
A. 2,009,500,000
B. 2,090,500,000
C. 2,900,050,000
D. 2,095,000,000
- 18.** The number that is 500,000 more than 1,295,300,000 is
A. 1,295,350,000
B. 1,300,300,000
C. 1,295,800,000
D. 1,296,300,000
- 20.** The standard form of 1,000 is
A. 10×10^2
B. 1×10^2
C. 0.1×10^4
D. 1×10^3
- 22.** The standard form of 375 is
A. 37.5×10
B. 3.75×10^3
C. 3.75×10^2
D. 0.375×10^3
- 24.** The standard form of 100 is
A. 1×10^2
B. 1×10^3
C. 10×10
D. 0.1×10^3
- 26.** The standard form of 1 is
A. 1×10
B. 0.1×10
C. 1×10^0
D. 10×10

- 27.** Which of the following is NOT in standard form?
- A. 3.5×10^4
 - B. 35×10^3
 - C. 1.2×10^6
 - D. 9.99×10^2
- 29.** 56,734 expressed correct to two significant figures is
- A. 56,000
 - B. 56,700
 - C. 57,000
 - D. 60,000
- 31.** 975.8674 expressed correct to three decimal places is
- A. 975.868
 - B. 975.86
 - C. 975.87
 - D. 975.867
- 33.** The number of significant figures in 0.04506 is
- A. 3
 - B. 5
 - C. 4
 - D. 6
- 35.** 3,456,789 expressed correct to one significant figure is
- A. 3,400,000
 - B. 3,500,000
 - C. 4,000,000
 - D. 3,000,000
- 37.** The population of Ghana in a certain year was 30,832,019. Written in words, this is
- A. three million, eight hundred and thirty-two thousand and nineteen
 - B. thirty million, eight hundred and twenty-three thousand and nineteen
 - C. thirty million, eight hundred and thirty-two thousand and nineteen
 - D. thirty billion, eight hundred and thirty-two thousand and nineteen
- 39.** The digit in the ten-thousands place of 3,764,528 is replaced by 0. The new number is
- 28.** The standard form of 10 is
- A. 1×10
 - B. 10×10^2
 - C. 0.1×10^2
 - D. 1×10^2
- 30.** 975.8674 expressed correct to two decimal places is
- A. 975.86
 - B. 975.87
 - C. 975.9
 - D. 976
- 32.** 56,734 expressed correct to three significant figures is
- A. 56,700
 - B. 57,000
 - C. 56,000
 - D. 56,730
- 34.** 48,296 rounded to the nearest thousand is
- A. 49,000
 - B. 48,000
 - C. 48,300
 - D. 50,000
- 36.** Adom earns GH¢2,500 a month after tax and his elder brother Arko earns three times as much. If there is no increase in their earnings, their total income after five years is
- A. GH¢600,000
 - B. GH¢150,000
 - C. GH¢300,000
 - D. GH¢60,000
- 38.** A company makes a profit of GH¢1,250,000 in a year. Written in standard form, this profit is
- A. 12.5×10^5
 - B. 1.25×10^6
 - C. 1.25×10^5
 - D. 0.125×10^7

- A. 3,764,028
- B. 3,760,528
- C. 3,064,528
- D. 3,704,528

- 40.** The distance from Accra to Tamale is about 615 km. A bus makes the journey 1,000 times in a year. The total distance travelled in a year, in standard form, is
- A. 6.15×10^5 km
 - B. 6.15×10^6 km
 - C. 61.5×10^4 km
 - D. 6.15×10^3 km

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Reading, writing and place value.

Question 2

(a) Skip counting.

Question 3

(a) Comparing and ordering.

Question 4

(a) Standard form.

Question 5

(a) Significant figures and decimal places.

Question 6

(a) Real-life problems.

Chapter 10: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The set of factors of 36 is
 - A. {1, 2, 3, 6, 12, 18, 36}
 - B. {2, 3, 4, 6, 9, 12, 18, 36}
 - C. {1, 2, 3, 4, 6, 9, 12, 18, 36}
 - D. {1, 2, 3, 4, 6, 8, 9, 12, 18, 36}
2. The number of elements in the set of factors of 42 is
 - A. 8
 - B. 6
 - C. 7
 - D. 9
3. The common factors of 42 and 36 are
 - A. {1, 2, 3}
 - B. {1, 3, 6, 9}
 - C. {2, 3, 6, 7}
 - D. {1, 2, 3, 6}
4. The highest common factor (HCF) of 42 and 36 is
 - A. 3
 - B. 6
 - C. 12
 - D. 2
5. Which of the following numbers is a factor of every whole number?
 - A. 1
 - B. 2
 - C. 3
 - D. 10
6. The set of common factors of 24 and 32 is
 - A. {1, 2, 4}
 - B. {1, 2, 4, 8}
 - C. {2, 4, 8, 16}
 - D. {1, 2, 3, 4, 6, 8}
7. The largest factor of 120 which is also a factor of 45 is
 - A. 5
 - B. 9
 - C. 15
 - D. 45
8. Which of the following numbers is a perfect square?
 - A. 144
 - B. 140
 - C. 48
 - D. 72
9. Which of the following numbers is NOT a perfect square?
 - A. 81
 - B. 100
 - C. 121
 - D. 110
10. The perfect squares among the numbers 16, 24, 32, 36, 48 and 64 are
 - A. 16, 24 and 36
 - B. 16, 32 and 64
 - C. 16, 36 and 64
 - D. 24, 36 and 48
11. The smallest perfect square greater than 100 is
 - A. 110
 - B. 121
 - C. 144
 - D. 101
12. A perfect square is a number which
 - A. has a whole number as its square root
 - B. is always even
 - C. has exactly two factors
 - D. is divisible by 4

- 13.** Which of the following sets contains only perfect squares?
A. {4, 9, 18, 25}
B. {1, 4, 8, 16}
C. {1, 4, 9, 16}
D. {9, 16, 27, 36}
- 15.** $\sqrt{81} =$
A. 8
B. 9
C. 18
D. 4.5
- 17.** To find $\sqrt{49}$, consecutive odd numbers starting from 1 are subtracted from 49 until the remainder is zero. The number of subtractions gives $\sqrt{49}$, which is
A. 5
B. 6
C. 49
D. 7
- 19.** $\sqrt{(36 \times 49)} =$
A. 85
B. 42
C. 1,764
D. 12
- 21.** A square plot of land has an area of 10,000 m². The length of each side of the plot is
A. 100 m
B. 1,000 m
C. 5,000 m
D. 50 m
- 23.** The set of elements common to two sets is called their
A. union
B. intersection
C. complement
D. universal set
- 25.** If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A \cap B$ is
A. {1, 2, 5, 6}
B. {1, 2, 3, 4, 5, 6}
C. {1, 2}
D. {3, 4}
- 27.** If two sets have no element in common, then their intersection is
A. the union of the two sets
- 14.** The next perfect square after 49 is
A. 56
B. 54
C. 50
D. 64
- 16.** $\sqrt{169} =$
A. 13
B. 17
C. 12
D. 15
- 18.** $\sqrt{400} =$
A. 40
B. 200
C. 20
D. 4
- 20.** The value of $\sqrt{225} + \sqrt{64}$ is
A. 17
B. 289
C. 15
D. 23
- 22.** Which of the following numbers has a whole number as its square root?
A. 90
B. 150
C. 196
D. 200
- 24.** The union of two sets A and B is the set of
A. all elements in A or B or both
B. elements common to A and B
C. elements in A but not in B
D. elements in neither A nor B
- 26.** If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A \cup B$ is
A. {3, 4}
B. {1, 2, 5, 6}
C. {1, 2, 3, 4, 5, 6}
D. {1, 2, 3, 4}

- B. the empty set $\{\}$
- C. one of the sets
- D. the universal set

- 29.** If $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 6, 9, 12\}$, then $A \cap B$ is
- A. $\{3, 6\}$
 - B. $\{2, 4, 8, 10\}$
 - C. $\{6\}$
 - D. $\{\}$

- 28.** The symbol $\cap$ represents
- A. intersection
 - B. union
 - C. subset
 - D. complement

- 31.** In a class, M is the set of students who like Mathematics and E the set of students who like English. The students who like both subjects are represented by
- $M \cup E$
 - $E - M$
 - $M - E$
 - $M \cap E$
- 33.** In a Venn diagram of two sets A and B, the region where the two circles overlap represents
- $A \cup B$
 - the elements in A only
 - $A \cap B$
 - the elements in neither set
- 35.** In a class of 40 students, 25 play football, 20 play volleyball and every student plays at least one of the games. The number of students who play both games is
- 10
 - 15
 - 20
 - 5
- 37.** In a survey of 60 people, 35 own a radio, 30 own a television and 10 own neither. The number of people who own both a radio and a television is
- 10
 - 25
 - 15
 - 20
- 39.** If $P = \{\text{factors of } 12\}$ and $Q = \{\text{factors of } 18\}$, the number of elements in $P \cup Q$ is
- 14
 - 6
 - 4
 - 8
- 30.** The number of elements in $A \cup B$, if $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 6, 9, 12\}$, is
- 9
 - 8
 - 7
 - 6
- 32.** If $n(A) = 12$, $n(B) = 15$ and $n(A \cap B) = 5$, then $n(A \cup B)$ is
- 22
 - 27
 - 17
 - 32
- 34.** In a village, 50 farmers grow maize and 60 grow rice. If all the 80 farmers grow at least one of the two crops, the number of farmers who grow both crops is
- 10
 - 30
 - 50
 - 20
- 36.** In a class of 40 students, 25 play football, 20 play volleyball and 5 play both games. The number of students who play football only is
- 20
 - 25
 - 15
 - 30
- 38.** If $P = \{\text{factors of } 12\}$ and $Q = \{\text{factors of } 18\}$, then $P \cap Q$ is
- $\{1, 2, 3\}$
 - $\{1, 2, 3, 6\}$
 - $\{2, 3, 4, 6\}$
 - $\{1, 6\}$
- 40.** In a market survey of 100 traders, 62 sell cloth, 48 sell shoes and 10 sell neither. The number of traders who sell both cloth and shoes is
- 20
 - 10
 - 30
 - 12

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Sets of factors and common factors.

Question 2

(a) Perfect squares.

Question 3

(a) Square roots.

Question 4

(a) Union and intersection of sets.

Question 5

(a) Venn diagrams and real-life problems.

Question 6

(a) Applied set problems.

Chapter 11: Ch3

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. $3.45 \times 100 =$
A. 34.5
B. 3,450
C. 345
D. 0.0345
2. $0.078 \times 1,000 =$
A. 78
B. 7.8
C. 780
D. 0.78
3. $6,250 \div 100 =$
A. 625
B. 6.25
C. 0.625
D. 62.5
4. $9.6 \div 10 =$
A. 96
B. 0.96
C. 0.096
D. 960
5. The benchmark fraction $\frac{1}{4}$ written as a decimal is
A. 0.25
B. 0.4
C. 0.14
D. 0.75
6. The decimal 0.2 written as a percentage is
A. 2%
B. 20%
C. 0.2%
D. 200%
7. 25% written as a benchmark fraction is
A. $\frac{1}{2}$
B. $\frac{1}{3}$
C. $\frac{1}{4}$
D. $\frac{3}{4}$
8. Using the halving-and-doubling strategy, 24×35 can be written as $12 \times \square$. The missing number is
A. 70
B. 17.5
C. 60
D. 84
9. Using the halving-and-doubling strategy, the value of 24×35 is
A. 420
B. 720
C. 980
D. 840
10. The value of 56×25 , computed mentally, is
A. 1,250
B. 1,500
C. 1,400
D. 1,450
11. Recall: $12 \times 12 =$
A. 124
B. 144
C. 132
D. 148
12. Given that $7 \times 8 = 56$, the value of 70×80 is
A. 5,600
B. 560
C. 56,000
D. 56
13. A basket of oranges costs GH¢4.50. The cost of 20 such baskets, computed mentally, is
A. GH¢45
B. GH¢95

- C. GH¢90
- D. GH¢80

15. $3,627.6 + 854.13 =$

- A. 4,471.73
- B. 4,481.73
- C. 4,481.63
- D. 4,381.73

17. $100,000 - 45,678 =$

- A. 55,322
- B. 54,422
- C. 64,322
- D. 54,322

19. $5.006 + 0.994 =$

- A. 5.996
- B. 6
- C. 6.4
- D. 5.1

21. The sum of 10,001, 20,002 and 30,003 is

- A. 60,006
- B. 60,060
- C. 60,600
- D. 66,000

23. $382 \times 856 =$

- A. 326,929
- B. 326,992
- C. 327,992
- D. 336,992

25. $12,096 \div 24 =$

- A. 404
- B. 54
- C. 506
- D. 504

27. Which of the following numbers is divisible by 9?

- A. 4,372
- B. 5,346
- C. 7,108
- D. 2,438

29. $48,000 \div 300 =$

- A. 16
- B. 1,600
- C. 160
- D. 1,060

14. $896,854 + 76,329 =$

- A. 963,183
- B. 973,283
- C. 972,183
- D. 973,183

16. $193.6 - 37.85 =$

- A. 155.75
- B. 166.25
- C. 156.75
- D. 155.25

18. $24,506 + 3,999 + 75,494 =$

- A. 104,999
- B. 102,999
- C. 103,999
- D. 103,099

20. $48,000 - 12,345 - 7,655 =$

- A. 30,000
- B. 27,000
- C. 29,000
- D. 28,000

22. $657 \times 27 =$

- A. 17,379
- B. 17,793
- C. 17,739
- D. 18,739

24. Using the distributive property, $325 \times 15 =$

- A. 4,875
- B. 4,857
- C. 4,785
- D. 4,895

26. Which of the following numbers is divisible by 8?

- A. 3,422
- B. 4,318
- C. 4,312
- D. 3,214

28. If a number is divisible by 6, it must also be divisible by

- A. 2 and 3
- B. 4 and 3
- C. 2 and 9
- D. 3 and 5

- 31.** $234 \times 405 =$
A. 94,700
B. 93,770
C. 94,670
D. 94,770
- 33.** An amount of GH¢2,477.25 was shared equally among three children. Each child received
A. GH¢825.25
B. GH¢725.75
C. GH¢825.75
D. GH¢826.75
- 35.** Mrs Yaboi bought 25.25 metres of cloth for her five children. If they shared the material equally, each child received
A. 5.5 m
B. 5.25 m
C. 5.005 m
D. 5.05 m
- 37.** From the same school budget, GH¢4,250 was approved for mending cracks on the netball pitch and GH¢9,990 for restocking the computer laboratory. The amount approved for the computer laboratory was more than that for the netball pitch by
A. GH¢5,640
B. GH¢5,840
C. GH¢5,740
D. GH¢6,740
- 39.** 15.6 kg of rice is shared equally into 12 bags. The mass of rice in each bag is
A. 1.2 kg
B. 1.25 kg
C. 1.4 kg
D. 1.3 kg
- 30.** The remainder when 10,000 is divided by 32 is
A. 8
B. 16
C. 24
D. 0
- 32.** Kofi bought 8 notebooks at GH¢12.00 each and Ama bought 12 pens at GH¢5.00 each. The total amount they spent altogether was
A. GH¢156
B. GH¢146
C. GH¢166
D. GH¢96
- 34.** A pack of chocolate contained 250 bars. Adwoa took 90 bars and shared the rest equally among her four friends. Each friend received
A. 62 bars
B. 40 bars
C. 50 bars
D. 35 bars
- 36.** A school approved GH¢2,940 for painting the school building and GH¢8,740 for buying new cadet uniforms. The total amount for the two projects was
A. GH¢11,680
B. GH¢11,580
C. GH¢10,680
D. GH¢11,670
- 38.** A car travels at a constant speed of 72 km/h. The distance it travels in 4.5 hours is
A. 314 km
B. 324 km
C. 334 km
D. 320 km
- 40.** Abena earns GH¢1,250 a month. Her total income in $2\frac{1}{2}$ years is
A. GH¢37,500
B. GH¢30,000
C. GH¢37,050
D. GH¢36,250

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Powers of 10 and benchmark fractions.

Question 2

(a) Mental mathematics strategies.

Question 3

(a) Addition and subtraction.

Question 4

(a) Multiplication and division.

Question 5

(a) Divisibility tests.

Question 6

(a) Story problems.

Chapter 12: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. According to the laws of indices, $a^m \times a^n =$
 - A. a^{mn}
 - B. a^{m-n}
 - C. a^{m+n}
 - D. $(a^m)^n$
2. $a^m \div a^n =$
 - A. a^{m-n}
 - B. a^{m+n}
 - C. $a^{m/n}$
 - D. a^{n-m}
3. $(a^m)^n =$
 - A. a^{m+n}
 - B. $a^{m/n}$
 - C. a^{m-n}
 - D. a^{mn}
4. $(ab)^n =$
 - A. ab^n
 - B. $a^n b^n$
 - C. $a^n + b^n$
 - D. $a^n b$
5. For any non-zero number a , the value of a^0 is
 - A. 1
 - B. 0
 - C. a
 - D. undefined
6. Written with a positive index, $a^{-n} =$
 - A. $-a^n$
 - B. $1/a^n$
 - C. a/n
 - D. n/a
7. Which of the following states Law 3 of indices?
 - A. $a^m \times a^n = a^{m+n}$
 - B. $a^m \div a^n = a^{m-n}$
 - C. $(a^m)^n = a^{mn}$
 - D. $(ab)^n = a^n b^n$
8. The index form of $4 \times 4 \times 4 \times 4 \times 4$ is
 - A. 4^5
 - B. 5^4
 - C. 4^4
 - D. 20
9. $2^5 \times 2^3 =$
 - A. 2^{15}
 - B. 4^8
 - C. 2^2
 - D. 2^8
10. $3^7 \div 3^4 =$
 - A. 3^{11}
 - B. 1^3
 - C. 3^3
 - D. 3^{28}
11. $(5^2)^3 =$
 - A. 5^5
 - B. 15,625
 - C. 3,125
 - D. 25,625
12. $(2 \times 3)^4 =$
 - A. 1,296
 - B. 24
 - C. 1,269
 - D. 48
13. $5 \times 16^2 =$
 - A. 1,208
 - B. 5,120
 - C. 1,280

D. 12,800

15. The fraction $27/32$ written in index form with prime bases is

- A. $3^3/2^4$
- B. $2^5/3^3$
- C. $3^3 \times 2^5$
- D. $3^3/2^5$

14. Evaluate $(16/81)^{-3/4}$

- A. $8/27$
- B. $27/8$
- C. $-27/8$
- D. $3/2$

17. Simplify: $(3^2 \times 3^5) \div 3^4$

- A. 3^{11}
- B. 9
- C. 27
- D. 81

19. Evaluate: 5^{-2}

- A. -25
- B. $1/10$
- C. -10
- D. $1/25$

21. Solve: $25 = 5^{2x}$

- A. 5
- B. 2
- C. 1
- D. 25

23. Solve: $2^5/2^3 = 2^{2x}$

- A. 2
- B. 4
- C. $1/2$
- D. 1

25. Solve: $3^{2x+1} = 27$

- A. 2
- B. 3
- C. 1
- D. 4

27. If $2^x = 32$, then $2^{x+1} =$

- A. 33
- B. 16
- C. 128
- D. 64

29. If $5^x = 1$, then $x =$

- A. 1
- B. 5
- C. 0
- D. -1

31. Saratu traces her ancestors back 12 generations. The number of ancestors she had in the 12th generation is

- A. 2,048
- B. 8,192
- C. 24
- D. 4,096

16. Simplify: $7^5 \div 7^2$

- A. 343
- B. 7^7
- C. 49
- D. 5

18. Simplify: $(2^3)^2 \times 2^4$

- A. 2^9
- B. 1,024
- C. 2^6
- D. 1,204

20. Simplify: $a^3 \times a^{-5}$

- A. $1/a^2$
- B. a^8
- C. a^2
- D. $1/a^8$

22. Solve: $2^{x+2} = 16$

- A. 4
- B. 2
- C. 6
- D. 8

24. Solve: $1/27 = 3^x$

- A. -3
- B. 3
- C. -9
- D. 9

26. Solve: $4^x = 64$

- A. 2
- B. 3
- C. 4
- D. 16

28. Solve: $9^x = 27$

- A. $3/2$
- B. 3
- C. 2
- D. 1

30. Solve: $10^x = 1,000,000$

- A. 5
- B. 6
- C. 7
- D. 10

32. The total number of Saratu's ancestors in the past 12 generations is

- A. 8,190
- B. 4,096
- C. 8,192

D. 4,094

- 33.** The number of ancestors in generation n is given by
- A. $2n$
 - B. n^2

- C. 2^n
D. $2 + n$
- 35.** A sheet of paper is folded in half four times. The number of layers of paper formed is
A. 8
B. 4
C. 12
D. 16
- 37.** The volume of a cube of side 2^3 cm is
A. 24 cm^3
B. 64 cm^3
C. 512 cm^3
D. 216 cm^3
- 39.** A message is shared so that each person shares it with 3 new people. The number of new people who receive it in the 4th round is
A. 27
B. 12
C. 64
D. 81
- 34.** A bacterium divides into two every hour. Starting with one bacterium, the number of bacteria after 6 hours is
A. 12
B. 64
C. 32
D. 62
- 36.** The area of a square of side 3^4 cm is
A. $6,561 \text{ cm}^2$
B. 81 cm^2
C. 324 cm^2
D. $2,187 \text{ cm}^2$
- 38.** Kofi's investment of GH¢50 doubles in value every year. Its value after 3 years is
A. GH¢150
B. GH¢400
C. GH¢300
D. GH¢800
- 40.** The population of a town doubles every 10 years. If the population is now 5,000, it will be 40,000 after
A. 30 years
B. 20 years
C. 40 years
D. 80 years

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The laws of indices.

Question 2

- (a) Evaluating powers.

Question 3

- (a) Exponential equations.

Question 4

- (a) Simplifying with the laws of indices.

Question 5

- (a) Saratu's ancestors.

Question 6

- (a) Real-life applications.

Chapter 13: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The fraction $\frac{6}{10}$ expressed in its simplest form is
 - A. $\frac{6}{5}$
 - B. $\frac{2}{5}$
 - C. $\frac{3}{10}$
 - D. $\frac{3}{5}$
2. Which of the following fractions is equivalent to $\frac{2}{3}$?
 - A. $\frac{8}{12}$
 - B. $\frac{6}{8}$
 - C. $\frac{4}{5}$
 - D. $\frac{9}{12}$
3. The fraction $\frac{12}{5}$ written as a mixed number is
 - A. $2 \frac{1}{5}$
 - B. $3 \frac{2}{5}$
 - C. $2 \frac{2}{5}$
 - D. $2 \frac{3}{5}$
4. The mixed number $2 \frac{5}{9}$ written as an improper fraction is
 - A. $\frac{19}{9}$
 - B. $\frac{23}{9}$
 - C. $\frac{25}{9}$
 - D. $\frac{14}{9}$
5. $\frac{3}{4}$ of 60 is
 - A. 40
 - B. 20
 - C. 15
 - D. 45
6. Which of the following fractions is the greatest?
 - A. $\frac{5}{6}$
 - B. $\frac{3}{4}$
 - C. $\frac{2}{3}$
 - D. $\frac{7}{10}$
7. The fraction $\frac{13}{8}$ written as a mixed number is
 - A. $1 \frac{3}{8}$
 - B. $2 \frac{5}{8}$
 - C. $1 \frac{5}{8}$
 - D. $1 \frac{6}{8}$
8. The shaded portion of a shape is 0.75. Written as a fraction in its simplest form, this is
 - A. $\frac{7}{5}$
 - B. $\frac{3}{4}$
 - C. $\frac{75}{100}$
 - D. $\frac{4}{3}$
9. $\frac{3}{4} + \frac{7}{8} =$
 - A. $\frac{10}{12}$
 - B. $1 \frac{1}{8}$
 - C. $\frac{13}{16}$
 - D. $1 \frac{5}{8}$
10. $\frac{4}{5} - \frac{1}{6} =$
 - A. $\frac{19}{30}$
 - B. $\frac{3}{30}$
 - C. $\frac{19}{11}$
 - D. $\frac{5}{11}$
11. $\frac{2}{3} \times \frac{3}{4} =$
 - A. $\frac{6}{7}$
 - B. $\frac{5}{12}$
 - C. $\frac{1}{2}$
 - D. $\frac{8}{9}$
12. $\frac{5}{8} \div 2 \frac{1}{2} =$
 - A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $2 \frac{1}{2}$
 - D. $\frac{25}{16}$

13. $1\frac{1}{2} + 2\frac{3}{4} =$

- A. $3\frac{4}{6}$
- B. $3\frac{1}{4}$
- C. $4\frac{1}{2}$
- D. $4\frac{1}{4}$

15. $\frac{2}{5} \times \frac{3}{7} =$

- A. $\frac{5}{12}$
- B. $\frac{6}{12}$
- C. $\frac{6}{35}$
- D. $\frac{5}{35}$

17. The reciprocal of $\frac{5}{9}$ is

- A. $-\frac{5}{9}$
- B. $\frac{5}{9}$
- C. $1\frac{5}{9}$
- D. $\frac{9}{5}$

19. Using the order of operations, $\frac{3}{4} + \frac{5}{8} \times \frac{4}{5} - \frac{1}{6} =$

- A. $1\frac{5}{12}$
- B. $\frac{11}{12}$
- C. $1\frac{1}{12}$
- D. $\frac{13}{24}$

21. Using the order of operations, $(\frac{3}{4} + \frac{5}{8}) \times \frac{4}{11} - \frac{1}{2} =$

- A. 0
- B. $\frac{1}{2}$
- C. 1
- D. $\frac{1}{4}$

23. $(\frac{5}{6} - \frac{1}{3}) \div \frac{1}{2} =$

- A. $\frac{1}{4}$
- B. 2
- C. 1
- D. $\frac{3}{4}$

25. $(2\frac{1}{2} - \frac{1}{2}) \times \frac{3}{4} =$

- A. $1\frac{1}{2}$
- B. $\frac{3}{4}$
- C. 1
- D. $2\frac{1}{4}$

27. Faako answered 42 out of 60 questions correctly. The percentage of her answers that were correct is

- A. 60%
- B. 70%
- C. 42%
- D. 75%

14. $3\frac{1}{2} - 1\frac{3}{4} =$

- A. $1\frac{3}{4}$
- B. $2\frac{1}{4}$
- C. $1\frac{1}{4}$
- D. $2\frac{3}{4}$

16. $\frac{3}{4} \div \frac{1}{8} =$

- A. $\frac{3}{32}$
- B. 6
- C. $\frac{1}{6}$
- D. 24

18. $\frac{7}{12} + \frac{1}{4} =$

- A. $\frac{5}{6}$
- B. $\frac{8}{16}$
- C. $\frac{7}{48}$
- D. $\frac{2}{3}$

20. Using the order of operations, $\frac{3}{4} \div \frac{3}{8} + (\frac{4}{5} - \frac{1}{2}) =$

- A. $2\frac{1}{5}$
- B. $2\frac{3}{10}$
- C. $1\frac{7}{10}$
- D. $2\frac{1}{2}$

22. $\frac{2}{3}$ of $(\frac{1}{2} + \frac{1}{6}) =$

- A. $\frac{2}{9}$
- B. $\frac{1}{3}$
- C. $\frac{2}{3}$
- D. $\frac{4}{9}$

24. $\frac{1}{2} + \frac{1}{4} \times \frac{2}{3} =$

- A. $\frac{1}{2}$
- B. $\frac{2}{3}$
- C. $\frac{1}{3}$
- D. $\frac{5}{12}$

26. $1\frac{1}{4} \div \frac{1}{2} + \frac{1}{4} =$

- A. $1\frac{1}{2}$
- B. 2
- C. $2\frac{3}{4}$
- D. $2\frac{1}{2}$

28. In a school, $\frac{2}{3}$ of the students eat from the school feeding programme and $\frac{1}{4}$ bring packed lunch. The fraction of the students who go home for lunch is

- A. $\frac{1}{12}$
- B. $\frac{5}{12}$
- C. $\frac{1}{7}$
- D. $\frac{11}{12}$

- 29.** Esi's orange drink is made with $\frac{2}{7}$ orange squash and Fusena's with $\frac{1}{4}$ orange squash. The drink that tastes stronger of orange is
- A. both taste equally strong
 - B. neither, since both are less than a half
 - C. Fusena's, because $\frac{1}{4}$ is greater than $\frac{2}{7}$
 - D. Esi's, because $\frac{2}{7}$ is greater than $\frac{1}{4}$
- 31.** The area of a rectangle whose sides measure $1\frac{1}{3}$ cm and $3\frac{3}{4}$ cm is
- A. 4 cm^2
 - B. 5 cm^2
 - C. 6 cm^2
 - D. $4\frac{1}{2}\text{ cm}^2$
- 33.** A rope $12\frac{1}{2}$ m long is cut into 5 equal pieces. The length of each piece is
- A. $2\frac{1}{2}$ m
 - B. $2\frac{1}{4}$ m
 - C. 2 m
 - D. 3 m
- 35.** $\frac{3}{5}$ of the 45 students in a class are girls. The number of boys in the class is
- A. 27
 - B. 15
 - C. 30
 - D. 18
- 37.** Kofi read $\frac{1}{4}$ of a 240-page book on Monday and $\frac{1}{3}$ of it on Tuesday. The total number of pages he read on the two days is
- A. 100
 - B. 120
 - C. 140
 - D. 160
- 39.** A farmer uses $\frac{1}{5}$ of his land for maize and $\frac{1}{4}$ of the remainder for vegetables. The fraction of the whole land used for vegetables is
- A. $\frac{1}{9}$
 - B. $\frac{1}{20}$
 - C. $\frac{1}{4}$
 - D. $\frac{1}{5}$
- 30.** The perimeter of a rectangle whose sides measure $1\frac{1}{3}$ cm and $3\frac{3}{4}$ cm is
- A. 5 cm
 - B. $9\frac{1}{6}$ cm
 - C. $10\frac{1}{6}$ cm
 - D. $20\frac{1}{3}$ cm
- 32.** Ama had GH¢240.00 and spent $\frac{2}{3}$ of it. The amount she spent was
- A. GH¢120.00
 - B. GH¢80.00
 - C. GH¢160.00
 - D. GH¢180.00
- 34.** A tank is $\frac{3}{8}$ full of water. The fraction of the tank that is empty is
- A. $\frac{3}{8}$
 - B. $\frac{1}{8}$
 - C. $\frac{8}{5}$
 - D. $\frac{5}{8}$
- 36.** Written as a decimal and rounded to two decimal places, $\frac{2}{3}$ is
- A. 0.67
 - B. 0.66
 - C. 0.6
 - D. 0.7
- 38.** Kofi read $\frac{1}{4}$ of a 240-page book on Monday and $\frac{1}{3}$ of it on Tuesday. The number of pages left unread is
- A. 140
 - B. 100
 - C. 80
 - D. 60
- 40.** A drink is made by mixing 2 cups of orange squash with 5 cups of water. The fraction of the drink that is squash is
- A. $\frac{2}{7}$
 - B. $\frac{2}{5}$
 - C. $\frac{5}{7}$
 - D. $\frac{1}{2}$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Review of fractions.

Question 2

(a) Operations on fractions.

Question 3

(a) Order of operations (BODMAS).

Question 4

(a) Fractions of quantities.

Question 5

(a) Measurement with fractions.

Question 6

(a) Integrated problems.

Chapter 14: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. 4 km written in metres is
 - A. 400 m
 - B. 40 m
 - C. 40,000 m
 - D. 4,000 m
2. 250 cm written in metres is
 - A. 2.5 m
 - B. 25 m
 - C. 0.25 m
 - D. 2,500 m
3. 3.2 litres written in millilitres is
 - A. 320 mL
 - B. 32 mL
 - C. 3,200 mL
 - D. 32,000 mL
4. 1 hour 15 minutes written in minutes is
 - A. 115 minutes
 - B. 75 minutes
 - C. 65 minutes
 - D. 90 minutes
5. Using ratio reasoning, since 1 m = 100 cm, a measurement in metres is converted to centimetres by multiplying by
 - A. 1/100
 - B. 10
 - C. 1,000
 - D. 100
6. Agbo walks 4 km to school every day and uses 60 minutes. His speed is
 - A. 4 km/h
 - B. 240 km/h
 - C. 0.067 km/h
 - D. 1 km/h
7. Rukiya covers 4,200 m in 45 minutes. Her speed is
 - A. 4.2 km/h
 - B. 4.5 km/h
 - C. 5.6 km/h
 - D. 9.3 km/h
8. Agbo walks 4 km in 60 minutes and Rukiya covers 4,200 m in 45 minutes. The faster walker is
 - A. Agbo
 - B. Rukiya
 - C. both are equally fast
 - D. it cannot be determined
9. If it takes 7 hours to mow 4 lawns, then at the same rate, the number of lawns that could be mowed in 35 hours is
 - A. 12
 - B. 28
 - C. 35
 - D. 20
10. The rate of mowing, if 4 lawns are mowed in 7 hours, is
 - A. 4/7 of a lawn per hour
 - B. 7/4 of a lawn per hour
 - C. 28 lawns per hour
 - D. 4 lawns per hour
11. Salamatu burns 756 calories while drumming for 3 hours. The number of calories she burns per hour is
 - A. 2,268
 - B. 225

- C. 252
- D. 352

13. A car travels 240 km in 3 hours. Its constant speed is

- A. 720 km/h
- B. 60 km/h
- C. 90 km/h
- D. 80 km/h

12. Six exercise books cost GH¢42.00.

The cost of one exercise book is

- A. GH¢6.00
- B. GH¢7.00
- C. GH¢8.00
- D. GH¢252.00

- 15.** Three kilograms of rice cost GH¢27.00. The cost of 5 kg at the same rate is
A. GH¢36.00
B. GH¢40.50
C. GH¢45.00
D. GH¢54.00
- 17.** A farmer uses 4 bags of fertilizer on 5 acres of land. The number of bags needed for 20 acres is
A. 8
B. 12
C. 20
D. 16
- 19.** On a distance-time graph, a horizontal line means the object is
A. moving with increasing speed
B. moving at a constant speed
C. at rest
D. moving backwards
- 21.** A trader travels from Buduata to Adawso, a distance of 20 miles. After 60 minutes she stops at Assin, which is 8 miles from Buduata. Her speed during the first hour is
A. 8 miles/h
B. 20 miles/h
C. 12 miles/h
D. 0.13 miles/h
- 23.** On a distance-time graph, a steep straight line shows that the object is
A. at rest
B. slowing down
C. moving at a high constant speed
D. travelling a decreasing distance
- 25.** A bus travels at 60 km/h for $2\frac{1}{2}$ hours. The distance it covers is
A. 150 km
B. 120 km
C. 24 km
D. 160 km
- 14.** A car travels at a constant speed of 80 km/h. The time it takes to travel 400 km is
A. 5 hours
B. 4 hours
C. 320 hours
D. 6 hours
- 16.** A machine produces 1,200 bottles in 8 hours. The number of bottles produced per hour is
A. 120
B. 150
C. 160
D. 9,600
- 18.** Kofi runs 1,500 m in 5 minutes. His time per kilometre is
A. $3\frac{1}{3}$ minutes
B. 3 minutes
C. 4 minutes
D. 5 minutes
- 20.** The gradient of a distance-time graph represents
A. distance
B. speed
C. time
D. acceleration
- 22.** In the same journey, the trader covers the remaining 12 miles to Adawso in 24 minutes. Her speed during this part of the journey is
A. 12 miles/h
B. 24 miles/h
C. 0.5 miles/h
D. 30 miles/h
- 24.** A cyclist covers 15 km in 45 minutes. His speed in km/h is
A. 15 km/h
B. 20 km/h
C. 45 km/h
D. 11.25 km/h
- 26.** From a travel graph, Ama leaves home at 7:00 and walks 4 km to school, arriving at 8:00. Her average speed is
A. 1 km/h
B. 8 km/h
C. 4 km/h

D. 2 km/h

28. Which of the following tables shows a proportional relationship?

- A. x: 1, 2, 3 and y: 5, 10, 16
- B. x: 1, 2, 3 and y: 5, 10, 15
- C. x: 1, 2, 3 and y: 5, 11, 17
- D. x: 1, 2, 3 and y: 5, 9, 13

27. Two quantities are proportional if

- A. their sum is constant
- B. their ratio is constant
- C. their product is constant
- D. their difference changes

- 29.** A straight-line graph which does NOT pass through the origin represents
- a proportional relationship
 - a unit rate of zero
 - a non-proportional relationship
 - no relationship
- 31.** The equation $y = 3x$ represents a proportional relationship because
- it has a constant term
 - its graph is a straight line through the origin
 - y increases by 3 each time
 - x is always positive
- 33.** The ratios 3 : 4 and 9 : 12 are equivalent, so the two quantities are
- proportional
 - not proportional
 - equal
 - undefined
- 35.** The same ant travels $\frac{27}{8}$ of an inch in 2 minutes 15 seconds. The constant of proportionality is
- $\frac{1}{20}$
 - $\frac{1}{40}$
 - $\frac{27}{8}$
 - 135
- 37.** In the proportional relationship $y = kx$, $y = 36$ when $x = 4$. The value of k is
- 144
 - 32
 - 9
 - 40
- 39.** Five kilograms of oranges cost GH¢60.00. The constant of proportionality (cost per kilogram) is
- GH¢300.00
 - GH¢5.00
 - GH¢60.00
 - GH¢12.00
- 30.** Two pens cost GH¢6.00 and five pens cost GH¢15.00. This shows
- a non-proportional relationship
 - a constant sum
 - a proportional relationship
 - a decreasing pattern
- 32.** Which of the following equations represents a proportional relationship?
- $y = 2x + 1$
 - $y = x - 3$
 - $y = 5x$
 - $y = 4 - x$
- 34.** An ant travels $\frac{9}{8}$ of an inch in 45 seconds. The constant of proportionality, in inches per second, is
- $\frac{9}{8}$
 - 45
 - $\frac{1}{8}$
 - $\frac{1}{40}$
- 36.** A table shows distances of 40 km in 2 h, 60 km in 3 h and 100 km in 5 h. The constant of proportionality is
- 20 km/h
 - 80 km/h
 - 200 km/h
 - 5 km/h
- 38.** The constant of proportionality is also known as the
- constant sum
 - unit rate
 - y-intercept
 - total amount
- 40.** If $y = kx$ with $k = 12$, the value of y when $x = 7$ is
- 84
 - 19
 - 5
 - 12

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Unit conversion by ratio reasoning.

Question 2

- (a) Unit rate problems.

Question 3

- (a) Speed and distance–time graphs.

Question 4

- (a) Proportional relationships.

Question 5

- (a) Constant of proportionality.

Question 6

- (a) Integrated problems.

Chapter 15: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The gradient of a straight line is a measure of its
 - A. length
 - B. steepness
 - C. smoothness
 - D. distance from the origin
2. The formula for the gradient of the line through (x_1, y_1) and (x_2, y_2) is
 - A. $(x_2 - x_1)/(y_2 - y_1)$
 - B. $(y_2 + y_1)/(x_2 + x_1)$
 - C. $(x_2 - y_2)/(x_1 - y_1)$
 - D. $(y_2 - y_1)/(x_2 - x_1)$
3. A line which rises from left to right has a gradient which is
 - A. positive
 - B. negative
 - C. zero
 - D. undefined
4. The gradient of a horizontal line is
 - A. 1
 - B. undefined
 - C. 0
 - D. -1
5. In the equation $y = mx + c$, the letter m represents the
 - A. y-intercept
 - B. gradient
 - C. x-coordinate
 - D. origin
6. In the equation $y = mx + c$, the letter c represents the
 - A. gradient
 - B. x-intercept
 - C. constant gradient
 - D. y-intercept
7. The gradient of the line passing through $A(1, 1)$ and $B(7, 2)$ is
 - A. $1/6$
 - B. 6
 - C. $1/5$
 - D. $-1/6$
8. The gradient of the line passing through $P(-2, 4)$ and $Q(3, 5)$ is
 - A. 5
 - B. $1/3$
 - C. $1/5$
 - D. $-1/5$
9. The gradient of the line passing through $C(3, -2)$ and $D(-3, 4)$ is
 - A. 1
 - B. -1
 - C. 0
 - D. 6
10. The gradient of the line passing through $(0, 0)$ and $(4, 10)$ is
 - A. $2/5$
 - B. 40
 - C. 4
 - D. $5/2$
11. The gradient of the line passing through $(2, 5)$ and $(2, 9)$ is
 - A. undefined
 - B. 0
 - C. 1
 - D. 4
12. A line passes through $(1, 3)$ and has gradient 2. Which of the following points also lies on the line?
 - A. $(2, 6)$
 - B. $(3, 6)$
 - C. $(4, 9)$
 - D. $(5, 10)$

- 13.** The gradient of the line $y = 5x + 13$ is
A. 13
B. 5
C. 18
D. -5
- 15.** The gradient and the y-intercept of the line $y = -3x + 12$ are respectively
A. -3 and 12
B. 3 and 12
C. -3 and -12
D. 12 and -3
- 17.** For the line $y = 2x - 3$, the value of y when $x = 4$ is
A. 8
B. 5
C. 11
D. 2
- 19.** The equation of the line with gradient $-5/2$ and y-intercept $7/2$ is
A. $y = (-5/2)x + 7/2$
B. $y = (5/2)x + 7/2$
C. $y = (-5/2)x - 7/2$
D. $y = (7/2)x - 5/2$
- 21.** The equation of the line passing through $(5, 4)$ and $(-10, -2)$ is
A. $y = (2/5)x - 2$
B. $y = (2/5)x + 2$
C. $y = (5/2)x + 2$
D. $y = (-2/5)x + 2$
- 23.** The gradient and the y-intercept of the line $5x + 4y - 3 = 0$ are respectively
A. $-5/4$ and $3/4$
B. $5/4$ and $3/4$
C. $-5/4$ and $-3/4$
D. $4/5$ and 3
- 25.** The cost of meat in a market is given by $y = 20x$, where y is the cost in cedis and x the weight in kg. The cost of 3.5 kg of meat is
A. GH¢70
B. GH¢35
C. GH¢80
D. GH¢23.5
- 14.** The gradient of the line $2x - 8y + 3 = 0$ is
A. 2
B. -8
C. 3
D. $1/4$
- 16.** The equation of the line with gradient 2 and y-intercept -3 is
A. $y = -3x + 2$
B. $y = 2x + 3$
C. $y = 2x - 3$
D. $y = 3x - 2$
- 18.** The equation of the line with gradient $1/2$ and y-intercept 4 is
A. $y = 4x + 1/2$
B. $y = (1/2)x - 4$
C. $y = 2x + 4$
D. $y = (1/2)x + 4$
- 20.** Using the point-slope form, the equation of the line with gradient $2/3$ passing through $(3, -1)$ is
A. $y = (2/3)x + 3$
B. $y = (3/2)x - 3$
C. $y = (2/3)x - 3$
D. $y = (-2/3)x + 3$
- 22.** Written in the form $y = mx + a$, the equation $5x + 4y - 3 = 0$ becomes
A. $y = (5/4)x + 3/4$
B. $y = (-5/4)x - 3/4$
C. $y = (4/5)x + 3$
D. $y = (-5/4)x + 3/4$
- 24.** The equation of a straight line which passes through the origin with gradient 4 is
A. $y = x + 4$
B. $y = 4$
C. $y = 4x$
D. $y = 4 - x$
- 26.** Using the relation $y = 20x$, the weight of meat that can be bought with GH¢80 is
A. 1,600 kg
B. 4 kg
C. 16 kg
D. 0.25 kg

- 27.** The diameter of a Wawa tree is modelled by $y = (2/5)x + 10$, where y is the diameter in inches after x years. The diameter after 20 years is
- A. 8 inches
 - B. 14 inches
 - C. 18 inches
 - D. 50 inches
- 29.** For the relation $y = (2/5)x + 10$, the value of y when $x = 0$ is
- A. 10
 - B. 0
 - C. $2/5$
 - D. 2
- 31.** The table x : 0, 1, 2, 3 and y : 2, 5, 8, 11 represents the relation
- A. $y = 2x + 3$
 - B. $y = x + 2$
 - C. $y = 3x + 2$
 - D. $y = 3x - 2$
- 33.** The distance you walk is modelled by $d = (1/3)h$, where d is the distance in km and h the number of hours walked. The distance walked after 6 hours is
- A. 2 km
 - B. 18 km
 - C. 3 km
 - D. $1/2$ km
- 35.** At the same rate, if Nhyira charged GH¢175 for a portrait, the number of hours she used is
- A. 3 hours
 - B. $3\frac{1}{2}$ hours
 - C. 4 hours
 - D. 125 hours
- 37.** A taxi charges a flat rate of GH¢5 plus GH¢2 for every kilometre. The equation of the cost y cedis for a journey of x km is
- A. $y = 2x + 5$
 - B. $y = 5x + 2$
 - C. $y = 7x$
 - D. $y = 2x - 5$
- 39.** A worker saves GH¢30 every month, starting with GH¢100. The equation of her savings y cedis after x months
- 28.** The diameter of the Wawa tree, using the relation $y = (2/5)x + 10$, in 100 years' time will be
- A. 40 inches
 - B. 60 inches
 - C. 110 inches
 - D. 50 inches
- 30.** In the table of values for $y = 3x + 1$, the value of y when $x = 4$ is
- A. 12
 - B. 13
 - C. 7
 - D. 11
- 32.** The gradient of the line represented by the table x : 0, 1, 2, 3 and y : 2, 5, 8, 11 is
- A. 2
 - B. 5
 - C. 1
 - D. 3
- 34.** Nhyira charges GH¢50 for every hour she spends painting a portrait. Her charge for a portrait which takes 3 hours is
- A. GH¢53
 - B. GH¢100
 - C. GH¢150
 - D. GH¢300
- 36.** The number of hours Nhyira will require to paint a portrait costing GH¢300 is
- A. 3 hours
 - B. 5 hours
 - C. $4\frac{1}{2}$ hours
 - D. 6 hours
- 38.** Using the taxi fares above, the cost of a 10 km journey is
- A. GH¢20
 - B. GH¢50
 - C. GH¢25
 - D. GH¢15

is

- A. $y = 100x + 30$
- B. $y = 30x + 100$
- C. $y = 130x$
- D. $y = 30x - 100$

40. Using the savings equation above, the worker's savings after 6 months will be

- A. GH¢180
- B. GH¢310
- C. GH¢210

D. GH¢280

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The gradient of a straight line.

Question 2

- (a) Equations of straight lines.

Question 3

- (a) Point-slope form and rearranging equations.

Question 4

- (a) Tables of values and graphs.

Question 5

- (a) Real-life linear graphs.

Question 6

- (a) The Wawa tree.

Chapter 16: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. When $6(x + 3)$ is expanded, the result is
 - A. $6x + 3$
 - B. $9x$
 - C. $6x + 18$
 - D. $6x^2 + 3$
2. When $-5x(3x + 4)$ is expanded, the result is
 - A. $-15x^2 + 20x$
 - B. $-15x^2 - 20x$
 - C. $-8x^2 - 9x$
 - D. $15x^2 - 20x$
3. The expression $3(x + 4) - 2(x - 5)$, when simplified, is
 - A. $x - 2$
 - B. $x + 2$
 - C. $5x + 2$
 - D. $x + 22$
4. The expression $2(6 - 5x) - 3(2 + 2x) - 4(3x - 1)$, when simplified, is
 - A. $10 - 28x$
 - B. $10 + 28x$
 - C. $2 - 28x$
 - D. $22 - 4x$
5. The expression $8 - (4 - a) - (6 - a)$, when simplified, is
 - A. $2 - 2a$
 - B. $2a - 2$
 - C. $18 - 2a$
 - D. $2a + 2$
6. The expression $(m + o - g) - (m - o + g)$, when simplified, is
 - A. $2o - 2g$
 - B. $2m - 2g$
 - C. $-2o + 2g$
 - D. $2m + 2o$
7. When $4a(2a - 5)$ is expanded, the result is
 - A. $8a - 20a$
 - B. $8a^2 - 5$
 - C. $6a^2 - 20a$
 - D. $8a^2 - 20a$
8. When $-3(2 - 4y)$ is expanded, the result is
 - A. $-6 - 12y$
 - B. $6 - 12y$
 - C. $-6 + 12y$
 - D. $-6 + 12y^2$
9. $(a + 2)(a + 3) =$
 - A. $a^2 + 6$
 - B. $a^2 + 5a + 6$
 - C. $a^2 + 6a + 5$
 - D. $2a^2 + 5a + 6$
10. $(2x + y)(2x - y) =$
 - A. $4x^2 - y^2$
 - B. $4x^2 + y^2$
 - C. $2x^2 - y^2$
 - D. $4x^2 - 2xy - y^2$
11. $(3x - 2y)(3x + 2y) =$
 - A. $9x^2 + 4y^2$
 - B. $9x^2 - 12xy + 4y^2$
 - C. $6x^2 - 4y^2$
 - D. $9x^2 - 4y^2$
12. $(2x + 3)^2 =$
 - A. $4x^2 + 9$
 - B. $4x^2 + 6x + 9$
 - C. $4x^2 + 12x + 9$
 - D. $2x^2 + 12x + 9$

- 13.** $(a + 2)^2 =$
A. $a^2 + 4$
B. $a^2 + 4a + 4$
C. $a^2 + 2a + 4$
D. $2a^2 + 4$
- 15.** $(x + 5)(x - 3) =$
A. $x^2 - 15$
B. $x^2 + 8x - 15$
C. $x^2 - 2x + 15$
D. $x^2 + 2x - 15$
- 17.** The expression $(a - b)/3 + 3a/2$, when simplified, is
A. $(11a - 2b)/6$
B. $(4a - b)/5$
C. $(11a - 2b)/5$
D. $(2a - 2b + 9a)/5$
- 19.** The expression $5/6r - 3/4r$, when simplified, is
A. $2/2r$
B. $1/12r^2$
C. $1/2r$
D. $1/12r$
- 21.** The expression $(3a + 5b)/4 + (a + b)/8$, when simplified, is
A. $(7a + 11b)/8$
B. $(4a + 6b)/8$
C. $(7a + 11b)/12$
D. $(6a + 10b)/8$
- 23.** The product of $3a/4$ and $8a/9$ is
A. $11a/13$
B. $2a^2/3$
C. $3a^2/2$
D. $2a/3$
- 25.** If $x = 2$, the value of $3/(x + 1) - 2/(x - 1)$ is
A. -1
B. 1
C. 5
D. -5
- 27.** If $x = 2$, $y = -2$, $z = 3$ and $a = 1$, the value of $5x + 7z^2 - 4a + 3y^2$ is
A. 63
B. 81
C. 45
D. 87
- 14.** $(x - 2)^2 =$
A. $x^2 - 4x + 4$
B. $x^2 - 4$
C. $x^2 + 4$
D. $x^2 - 2x + 4$
- 16.** $(p + q)(p - q) =$
A. $p^2 + q^2$
B. $p^2 - q^2$
C. $p^2 - 2pq + q^2$
D. $2p - 2q$
- 18.** The expression $2a/3 - (a - b)/2$, when simplified, is
A. $(a - 3b)/6$
B. $(4a - a - b)/6$
C. $(a + 3b)/6$
D. $(3a + 2b)/6$
- 20.** The expression $(2x - 1)/3 - (x + 3)/2$, when simplified, is
A. $(x + 2)/6$
B. $(x - 11)/6$
C. $(x - 11)/5$
D. $(3x - 4)/6$
- 22.** The expression $x/3 + x/6$, when simplified, is
A. $2x/9$
B. $x/9$
C. $x/2$
D. $2x/6^2$
- 24.** $10x/3$ divided by $5x/9$ is
A. $50x^2/27$
B. $3/2$
C. $2/3$
D. 6
- 26.** If $x = 2$, the value of $1/(x - 1) + 2/(x + 1)$ is
A. 3
B. 1
C. $2/3$
D. $5/3$
- 28.** If $x = 2$ and $y = -2$, the value of $x^2 + y^2$ is
A. 0
B. 4
C. 8
D. -8

- 29.** If $x = 2$, $y = -2$ and $z = 3$, the value of $2x + 3y - z$ is
A. -5
B. 5
C. 1
D. -1
- 31.** If $x = 2$ and $y = -2$, the value of $(12xy/7) \times (14x/20)$ is
A. $48/5$
B. $24/5$
C. $-48/5$
D. $-24/5$
- 33.** The expression $3ax + 6ay$ factorised completely is
A. $3(ax + 2y)$
B. $6a(x + y)$
C. $3a(x + 6y)$
D. $3a(x + 2y)$
- 35.** The expression $100x - 25x^2$ factorised completely is
A. $25(4x - x^2)$
B. $5x(20 - 5x)$
C. $25x(4 - x)$
D. $25x(x - 4)$
- 37.** The expression $x^2 + 5x + 6$ factorised is
A. $(x + 2)(x + 3)$
B. $(x + 1)(x + 6)$
C. $(x + 5)(x + 1)$
D. $(x - 2)(x - 3)$
- 39.** The expression $ax - ay - xy + y^2$ factorised by grouping is
A. $(x + y)(a - y)$
B. $(x - y)(a + y)$
C. $(x - a)(y - x)$
D. $(x - y)(a - y)$
- 30.** If $a = 1$ and $b = -1$, the value of $a^2 - b^2$ is
A. 2
B. -2
C. 1
D. 0
- 32.** If $x = 2$, the value of $3x^2 - 5x + 4$ is
A. 6
B. 10
C. 2
D. 16
- 34.** The expression $54 - 81x$ factorised completely is
A. $9(6 - 9x)$
B. $27(2 - 3x)$
C. $27(3 - 2x)$
D. $54(1 - 81x)$
- 36.** The expression $x^2 - 25$ factorised is
A. $(x - 5)^2$
B. $(x - 5)(x + 5)$
C. $(x - 25)(x + 1)$
D. $x(x - 25)$
- 38.** The expression $4x^2 - 9y^2$ factorised is
A. $(2x - 3y)^2$
B. $(4x + 9y)(x - y)$
C. $(2x - 3y)(2x + 3y)$
D. $4x(x - 9y)$
- 40.** The expression $3x^2 + 2xy - 12xz - 8yz$ factorised by grouping is
A. $(3x + 2y)(x + 4z)$
B. $(3x + 2y)(x - 4z)$
C. $(3x - 2y)(x - 4z)$
D. $(3x + 8y)(x - z)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Expanding brackets.

Question 2

(a) Multiplication of binomial expressions.

Question 3

(a) Operations on algebraic fractions.

Question 4

(a) Substitution and evaluation.

Question 5

(a) Factorisation.

Question 6

(a) Integrated algebra.

Chapter 17: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A whole number is less than 17. If the number is x , the inequality representing the statement is
 - A. $x < 17$
 - B. $x > 17$
 - C. $x \leq 17$
 - D. $x \geq 17$
2. Eight less than the product of -3 and a number x is greater than -26 . The inequality representing the statement is
 - A. $-3x + 8 > -26$
 - B. $-3(x - 8) > -26$
 - C. $-3x - 8 > -26$
 - D. $8 - 3x < -26$
3. Kwaakye's profit for March was GH¢32.00, which was at least GH¢12.00 less than his profit for February. If his February profit is p cedis, the inequality representing the situation is
 - A. $32 \leq p - 12$
 - B. $32 \geq p - 12$
 - C. $p \geq 32 - 12$
 - D. $32 > p + 12$
4. Ama is older than 12 years. If her age is a years, the inequality representing the statement is
 - A. $a < 12$
 - B. $a = 12$
 - C. $a \geq 12$
 - D. $a > 12$
5. The total cost of x books at GH¢5.00 each does not exceed GH¢100.00. The inequality representing the statement is
 - A. $5x \leq 100$
 - B. $5x \geq 100$
 - C. $5x < 100$
 - D. $x + 5 \leq 100$
6. The statement " x is at most 9" means
 - A. $x > 9$
 - B. $x < 9$
 - C. $x \leq 9$
 - D. $x \geq 9$
7. A lift can carry at most 600 kg. If x bags of rice, each weighing 50 kg, are put into the lift, the inequality representing the statement is
 - A. $50x \geq 600$
 - B. $50x \leq 600$
 - C. $x \leq 600$
 - D. $600x \leq 50$
8. Which of the following word statements matches the inequality $2x + 5 > 15$?
 - A. Twice the sum of a number and 5 is greater than 15
 - B. Five times a number, plus 2, is greater than 15
 - C. The sum of twice a number and 15 is greater than 5
 - D. Five more than twice a number is greater than 15
9. Solve the inequality $x - 5 > 1$
 - A. $x > 6$
 - B. $x > 4$
 - C. $x < 6$
 - D. $x > -4$
10. Solve the inequality $x - 3 \geq 2$
 - A. $x \geq 1$
 - B. $x \geq -1$
 - C. $x \leq 5$
 - D. $x \geq 5$

- 11.** Solve the inequality $13 - x < 12$
- A. $x < 1$
 - B. $x > 25$
 - C. $x > 1$
 - D. $x < 25$
- 12.** Solve the inequality $3x \leq 8 + x$
- A. $x \leq 8$
 - B. $x \leq 4$
 - C. $x \geq 4$
 - D. $x \leq 2$
- 13.** Solve the inequality $x + 7 \leq 15$
- A. $x \leq 8$
 - B. $x \geq 8$
 - C. $x \leq 22$
 - D. $x < 8$
- 14.** Solve the inequality $2x - 13 > 29$
- A. $x > 8$
 - B. $x > 16$
 - C. $x > 42$
 - D. $x > 21$
- 15.** Solve the inequality $4x + 3 \geq 23$
- A. $x \geq 4$
 - B. $x \geq 5$
 - C. $x \leq 5$
 - D. $x \geq 6$
- 16.** Solve the inequality $-2x < 10$
- A. $x > -5$
 - B. $x < -5$
 - C. $x > 5$
 - D. $x < 5$
- 17.** Solve the inequality $7 - 2x \geq 1$
- A. $x \geq 3$
 - B. $x \leq -3$
 - C. $x \leq 3$
 - D. $x \geq -3$
- 18.** Solve the inequality $x/4 > 3$
- A. $x > 3/4$
 - B. $x > 12$
 - C. $x < 12$
 - D. $x > 7$
- 19.** Solve the inequality $5x - 9 > -5$
- A. $x > -4/5$
 - B. $x > 14/5$
 - C. $x < 4/5$
 - D. $x > 4/5$
- 20.** Solve the inequality $6 - x \leq 2$
- A. $x \geq 4$
 - B. $x \leq 4$
 - C. $x \geq -4$
 - D. $x \leq 8$
- 21.** Solve the inequality $x + 4 \leq 3x - 16$
- A. $x \leq 10$
 - B. $x \geq 6$
 - C. $x \geq 10$
 - D. $x \leq 6$
- 22.** The smallest whole number which satisfies $2x + 1 > 9$ is
- A. 4
 - B. 5
 - C. 9
 - D. 10
- 23.** If $x < 4$ and x is a whole number, the solution set is
- A. $\{1, 2, 3\}$
 - B. $\{0, 1, 2, 3, 4\}$
 - C. $\{4, 3, 2, 1\}$
 - D. $\{0, 1, 2, 3\}$
- 24.** If $2x > 24$ and x is a whole number, then
- A. $x > 12$
 - B. $x < 12$
 - C. $x > 48$
 - D. $x \geq 12$
- 25.** If $x \geq 5$ and x is a whole number less than 9, the solution set is
- A. $\{5, 6, 7\}$
 - B. $\{5, 6, 7, 8\}$
 - C. $\{6, 7, 8, 9\}$
 - D. $\{5, 6, 7, 8, 9\}$
- 26.** The solution set of $x + 2 \leq 5$, where x is a whole number, is
- A. $\{0, 1, 2\}$
 - B. $\{1, 2, 3\}$
 - C. $\{0, 1, 2, 3\}$
 - D. $\{3, 2, 1, 0, -1\}$
- 27.** If $3x < 15$ and x is a positive whole number, the solution set is
- A. $\{1, 2, 3, 4\}$
 - B. $\{0, 1, 2, 3, 4\}$
 - C. $\{1, 2, 3, 4, 5\}$

D. $\{5, 6, 7, \dots\}$

- 29.** If $x - 1 > 2$ and x is a whole number at most 6, the solution set is
- $\{3, 4, 5, 6\}$
 - $\{4, 5, 6\}$
 - $\{4, 5\}$
 - $\{1, 2, 3\}$
- 31.** If $2x + 3 \leq 11$ and x is a whole number, the greatest possible value of x is
- 4
 - 3
 - 5
 - 7
- 33.** On a number line, the inequality $x \geq 3$ is represented by
- an open dot at 3 and an arrow to the right
 - a solid dot at 3 and an arrow to the right
 - a solid dot at 3 and an arrow to the left
 - an open dot at 3 and an arrow to the left
- 35.** A student must score at least 40 marks to pass a test. If Kofi scores x marks and passes, then
- $x > 40$
 - $x < 40$
 - $x \geq 40$
 - $x \leq 40$
- 37.** Adjoa has x oranges. She shares them equally among 5 friends and each friend receives at most 6 oranges. The inequality representing the statement is
- $x \leq 30$
 - $x \geq 30$
 - $x < 11$
 - $x \leq 6$
- 39.** A farmer has at most GH¢500.00 to spend on seedlings costing GH¢20.00 each. The greatest number of seedlings he can buy is
- 24
 - 25
 - 26
 - 50
- 28.** The inequality satisfied by all the numbers in the set $\{6, 7, 8, \dots\}$ is
- $x > 8$
 - $x \leq 6$
 - $x < 6$
 - $x \geq 6$
- 30.** The number of whole numbers which satisfy $x + 5 < 9$ is
- 3
 - 5
 - 4
 - 9
- 32.** The solution set of $x > -2$, where x is an integer between -4 and 3 , is
- $\{-3, -2, -1, 0\}$
 - $\{0, 1, 2\}$
 - $\{-2, -1, 0, 1, 2\}$
 - $\{-1, 0, 1, 2\}$
- 34.** On a number line, an open dot at 2 with shading to the right represents
- $x > 2$
 - $x \geq 2$
 - $x < 2$
 - $x \leq 2$
- 36.** A taxi can carry at most 4 passengers. If p passengers enter the taxi, the inequality representing the statement is
- $p > 4$
 - $p \geq 4$
 - $p < 4$
 - $p \leq 4$
- 38.** The solution set of $x \leq 2$, where x is a whole number, is shown on a number line by shading the points
- 2 only
 - 0 and 1 only
 - 0, 1 and 2
 - all numbers less than 2
- 40.** Amina's weight in kilograms, increased by 5, does not exceed 60. The greatest possible value of her weight w is
- 60
 - 65
 - 50
 - 55

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Translating word problems.

Question 2

(a) Solving linear inequalities.

Question 3

(a) Solution sets in given domains.

Question 4

(a) Number lines.

Question 5

(a) Real-life applications.

Question 6

(a) Integrated problems.

ANSWER KEY

Ch10**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Angle facts (straight line, point, vertica...	Remember
2	D	Angle facts (straight line, point, vertica...	Remember
3	A	Angle facts (straight line, point, vertica...	Understand
4	C	Angle facts (straight line, point, vertica...	Apply
5	B	Angle facts (straight line, point, vertica...	Apply
6	C	Angle facts (straight line, point, vertica...	Analyse
7	A	Angle facts (straight line, point, vertica...	Apply
8	D	Angle facts (straight line, point, vertica...	Apply
9	B	Transversals: alternate & corresponding an...	Remember
10	A	Transversals: alternate & corresponding an...	Understand
11	D	Transversals: alternate & corresponding an...	Understand
12	B	Transversals: alternate & corresponding an...	Remember
13	C	Transversals: alternate & corresponding an...	Apply
14	A	Transversals: alternate & corresponding an...	Analyse
15	D	Transversals: alternate & corresponding an...	Analyse
16	C	Transversals: alternate & corresponding an...	Apply
17	A	Transversals: alternate & corresponding an...	Analyse
18	B	Transversals: alternate & corresponding an...	Apply
19	D	Transversals: alternate & corresponding an...	Analyse
20	C	Transversals: alternate & corresponding an...	Understand
21	B	Angles in triangles	Remember
22	D	Angles in triangles	Apply
23	A	Angles in triangles	Apply
24	C	Angles in triangles	Remember
25	B	Angles in triangles	Apply
26	A	Angles in polygons	Remember
27	C	Angles in polygons	Apply
28	D	Angles in polygons	Apply
29	B	Angles in polygons	Apply
30	A	Angles in polygons	Remember
31	D	Angles in polygons	Apply
32	C	Angles in polygons	Analyse
33	A	Angles in polygons	Analyse
34	B	Angles in triangles	Apply
35	D	Angles in triangles	Apply
36	B	Angles in triangles	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Angles in triangles	Apply
38	A	Transversals: alternate & corresponding an...	Analyse
39	D	Angles in polygons	Analyse
40	B	Angles in polygons	Analyse

Answer Distribution:

A: 10 (25%) | B: 11 (27%) | C: 9 (22%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 10 (25%) | Apply: 18 (45%) | Remember: 8 (20%) | Understand: 4 (10%)

Ch11**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Construction & bisection of angles	Understand
2	A	Construction & bisection of angles	Understand
3	C	Construction & bisection of angles	Remember
4	D	Construction & bisection of angles	Apply
5	A	Construction & bisection of angles	Apply
6	D	Construction & bisection of angles	Understand
7	C	Construction & bisection of angles	Understand
8	B	Construction & bisection of angles	Understand
9	D	Construction & bisection of angles	Remember
10	A	Construction & bisection of angles	Analyse
11	D	Construction of triangles	Understand
12	C	Construction of triangles	Apply
13	B	Construction of triangles	Understand
14	A	Construction of triangles	Understand
15	C	Construction of triangles	Apply
16	D	Construction of triangles	Apply
17	B	Construction of triangles	Understand
18	A	Construction of triangles	Understand
19	D	Construction of triangles	Understand
20	B	Construction of triangles	Apply
21	C	Construction of triangles	Analyse
22	A	Construction of triangles	Analyse
23	D	Construction of triangles	Remember
24	B	Construction of triangles	Analyse
25	C	Loci under given conditions	Understand
26	A	Loci under given conditions	Understand
27	D	Loci under given conditions	Understand
28	C	Loci under given conditions	Understand
29	B	Loci under given conditions	Apply
30	A	Loci under given conditions	Apply
31	B	Loci under given conditions	Analyse
32	D	Loci under given conditions	Analyse
33	A	Loci under given conditions	Analyse
34	C	Loci under given conditions	Understand
35	B	Loci under given conditions	Apply
36	A	Loci under given conditions	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Loci under given conditions	Apply
38	D	Construction & bisection of angles	Apply
39	B	Construction of triangles	Understand
40	A	Loci under given conditions	Create

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 7 (17%) | Apply: 12 (30%) | Create: 1 (2%) | Remember: 3 (7%) | Understand: 17 (42%)

Ch12

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Area & circumference of circles	Remember
2	B	Area & circumference of circles	Apply
3	C	Area & circumference of circles	Apply
4	A	Area & circumference of circles	Apply
5	C	Area & circumference of circles	Apply
6	A	Area & circumference of circles	Understand
7	B	Area & circumference of circles	Understand
8	D	Area & circumference of circles	Apply
9	C	Area & circumference of circles	Apply
10	B	Area & circumference of circles	Analyse
11	C	Area & circumference of circles	Apply
12	A	Area & circumference of circles	Analyse
13	B	Pythagoras theorem problems	Remember
14	C	Pythagoras theorem problems	Apply
15	D	Pythagoras theorem problems	Apply
16	B	Pythagoras theorem problems	Apply
17	A	Pythagoras theorem problems	Apply
18	D	Pythagoras theorem problems	Apply
19	B	Pythagoras theorem problems	Apply
20	D	Pythagoras theorem problems	Apply
21	B	Pythagoras theorem problems	Apply
22	C	Pythagoras theorem problems	Apply
23	D	Pythagoras theorem problems	Apply
24	A	Pythagoras theorem problems	Apply
25	C	Pythagoras theorem problems	Understand
26	B	Pythagoras theorem problems	Analyse
27	D	Pythagoras theorem problems	Apply
28	A	Pythagoras theorem problems	Apply
29	C	Trigonometric ratios	Remember
30	B	Trigonometric ratios	Remember
31	C	Trigonometric ratios	Apply
32	D	Trigonometric ratios	Apply
33	A	Trigonometric ratios	Apply
34	B	Trigonometric ratios	Understand
35	C	Trigonometric ratios	Apply
36	A	Trigonometric ratios	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	B	Area & circumference of circles	Apply
38	D	Pythagoras theorem problems	Apply
39	C	Area & circumference of circles	Analyse
40	A	Pythagoras theorem problems	Apply

Answer Distribution:

A: 10 (25%) | B: 11 (27%) | C: 11 (27%) | D: 8 (20%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 28 (70%) | Remember: 4 (10%) | Understand: 4 (10%)

Ch13**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Vector concepts & notation	Remember
2	A	Vector concepts & notation	Remember
3	D	Vector concepts & notation	Remember
4	B	Vector concepts & notation	Apply
5	A	Vector concepts & notation	Understand
6	D	Vector concepts & notation	Understand
7	C	Addition of vectors	Apply
8	B	Addition of vectors	Apply
9	A	Addition of vectors	Apply
10	D	Addition of vectors	Apply
11	B	Addition of vectors	Apply
12	C	Addition of vectors	Analyse
13	A	Addition of vectors	Analyse
14	D	Addition of vectors	Apply
15	B	Subtraction & scalar multiplication	Apply
16	A	Subtraction & scalar multiplication	Apply
17	C	Subtraction & scalar multiplication	Apply
18	B	Subtraction & scalar multiplication	Apply
19	D	Subtraction & scalar multiplication	Apply
20	A	Subtraction & scalar multiplication	Apply
21	D	Subtraction & scalar multiplication	Analyse
22	B	Subtraction & scalar multiplication	Apply
23	A	Subtraction & scalar multiplication	Analyse
24	C	Subtraction & scalar multiplication	Apply
25	D	Subtraction & scalar multiplication	Understand
26	C	Subtraction & scalar multiplication	Apply
27	B	Vector equality & applications	Understand
28	A	Vector equality & applications	Apply
29	D	Vector equality & applications	Analyse
30	C	Vector equality & applications	Apply
31	A	Vector equality & applications	Apply
32	D	Vector equality & applications	Apply
33	B	Vector equality & applications	Apply
34	C	Vector equality & applications	Apply
35	A	Vector equality & applications	Analyse
36	B	Vector equality & applications	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	D	Vector equality & applications	Understand
38	C	Vector equality & applications	Apply
39	A	Vector equality & applications	Analyse
40	B	Vector equality & applications	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 8 (20%) | Apply: 23 (57%) | Remember: 3 (7%) | Understand: 6 (15%)

Ch14

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Rotation concepts & real-life situations	Understand
2	B	Rotation concepts & real-life situations	Remember
3	A	Rotation concepts & real-life situations	Remember
4	C	Rotation concepts & real-life situations	Understand
5	D	Rotation concepts & real-life situations	Remember
6	B	Rotation concepts & real-life situations	Remember
7	C	Rotation concepts & real-life situations	Remember
8	A	Rotation concepts & real-life situations	Understand
9	D	Rotation rules about the origin	Apply
10	B	Rotation rules about the origin	Apply
11	A	Rotation rules about the origin	Apply
12	D	Rotation rules about the origin	Apply
13	C	Rotation rules about the origin	Apply
14	B	Rotation rules about the origin	Apply
15	A	Rotation rules about the origin	Apply
16	C	Rotation rules about the origin	Understand
17	B	Rotation rules about the origin	Remember
18	D	Rotation rules about the origin	Remember
19	A	Rotation rules about the origin	Analyse
20	C	Rotation rules about the origin	Analyse
21	D	Rotation rules about the origin	Apply
22	A	Rotation rules about the origin	Understand
23	B	Rotation rules about the origin	Analyse
24	C	Rotation rules about the origin	Analyse
25	D	Angle of rotation	Understand
26	B	Angle of rotation	Apply
27	A	Angle of rotation	Apply
28	D	Angle of rotation	Apply
29	C	Angle of rotation	Understand
30	B	Angle of rotation	Apply
31	A	Angle of rotation	Apply
32	C	Angle of rotation	Apply
33	B	Congruence under rotation	Understand
34	D	Congruence under rotation	Understand
35	A	Congruence under rotation	Understand
36	C	Congruence under rotation	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	D	Congruence under rotation	Apply
38	A	Congruence under rotation	Apply
39	B	Congruence under rotation	Understand
40	C	Congruence under rotation	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 16 (40%) | Remember: 8 (20%) | Understand: 12 (30%)

Ch15**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Types of data	Remember
2	D	Types of data	Understand
3	B	Types of data	Understand
4	C	Types of data	Understand
5	A	Types of data	Remember
6	C	Types of data	Remember
7	B	Types of data	Understand
8	A	Types of data	Understand
9	D	Types of data	Analyse
10	C	Types of data	Understand
11	B	Methods of data collection	Apply
12	A	Methods of data collection	Apply
13	D	Methods of data collection	Apply
14	C	Methods of data collection	Understand
15	B	Methods of data collection	Remember
16	D	Methods of data collection	Analyse
17	C	Frequency tables & stem-and-leaf plots	Understand
18	B	Frequency tables & stem-and-leaf plots	Understand
19	A	Frequency tables & stem-and-leaf plots	Apply
20	D	Frequency tables & stem-and-leaf plots	Apply
21	A	Frequency tables & stem-and-leaf plots	Analyse
22	B	Frequency tables & stem-and-leaf plots	Analyse
23	D	Frequency tables & stem-and-leaf plots	Apply
24	C	Frequency tables & stem-and-leaf plots	Apply
25	A	Frequency tables & stem-and-leaf plots	Understand
26	C	Frequency tables & stem-and-leaf plots	Understand
27	A	Graphs, charts & pictograms	Understand
28	B	Graphs, charts & pictograms	Understand
29	D	Graphs, charts & pictograms	Remember
30	C	Graphs, charts & pictograms	Apply
31	B	Graphs, charts & pictograms	Apply
32	D	Graphs, charts & pictograms	Apply
33	C	Graphs, charts & pictograms	Apply
34	A	Graphs, charts & pictograms	Apply
35	B	Graphs, charts & pictograms	Apply
36	D	Graphs, charts & pictograms	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	A	Graphs, charts & pictograms	Apply
38	C	Graphs, charts & pictograms	Apply
39	B	Graphs, charts & pictograms	Understand
40	D	Graphs, charts & pictograms	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 16 (40%) | Remember: 5 (12%) | Understand: 15 (37%)

Ch16**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Mean of ungrouped data	Apply
2	C	Mean of ungrouped data	Apply
3	A	Mean of ungrouped data	Apply
4	D	Mean of ungrouped data	Analyse
5	B	Mean of ungrouped data	Analyse
6	A	Mean of ungrouped data	Apply
7	D	Mean of ungrouped data	Apply
8	C	Mean of ungrouped data	Apply
9	B	Mean of ungrouped data	Analyse
10	A	Mean of ungrouped data	Analyse
11	D	Median & mode	Apply
12	B	Median & mode	Apply
13	A	Median & mode	Apply
14	C	Median & mode	Analyse
15	D	Median & mode	Analyse
16	C	Median & mode	Apply
17	A	Median & mode	Apply
18	D	Median & mode	Apply
19	B	Median & mode	Analyse
20	C	Median & mode	Understand
21	A	Range	Apply
22	B	Range	Apply
23	C	Range	Analyse
24	D	Range	Understand
25	A	Range	Remember
26	D	Range	Apply
27	C	Choosing the appropriate measure	Understand
28	B	Choosing the appropriate measure	Apply
29	A	Choosing the appropriate measure	Apply
30	D	Choosing the appropriate measure	Understand
31	B	Choosing the appropriate measure	Apply
32	A	Choosing the appropriate measure	Apply
33	D	Choosing the appropriate measure	Analyse
34	C	Choosing the appropriate measure	Analyse
35	B	Integrated problems	Apply
36	C	Integrated problems	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	D	Integrated problems	Apply
38	A	Integrated problems	Apply
39	B	Integrated problems	Analyse
40	C	Integrated problems	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 11 (27%) | Apply: 23 (57%) | Remember: 1 (2%) | Understand: 5 (12%)

Ch17**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Sample spaces & independent events	Remember
2	A	Sample spaces & independent events	Understand
3	C	Sample spaces & independent events	Understand
4	B	Sample spaces & independent events	Apply
5	D	Sample spaces & independent events	Understand
6	B	Sample spaces & independent events	Apply
7	A	Sample spaces & independent events	Understand
8	D	Sample spaces & independent events	Understand
9	C	Sample spaces & independent events	Understand
10	B	Sample spaces & independent events	Apply
11	A	Sample spaces & independent events	Apply
12	C	Sample spaces & independent events	Apply
13	B	Probabilities of events	Apply
14	D	Probabilities of events	Apply
15	A	Probabilities of events	Apply
16	C	Probabilities of events	Apply
17	D	Probabilities of events	Apply
18	A	Probabilities of events	Analyse
19	B	Probabilities of events	Apply
20	C	Probabilities of events	Apply
21	D	Probabilities of events	Apply
22	B	Probabilities of events	Apply
23	A	Probabilities of events	Analyse
24	C	Probabilities of events	Remember
25	D	Probabilities of events	Remember
26	A	Probabilities of events	Apply
27	B	Fractions, decimals, percentages & ratios	Apply
28	D	Fractions, decimals, percentages & ratios	Apply
29	C	Fractions, decimals, percentages & ratios	Apply
30	A	Fractions, decimals, percentages & ratios	Apply
31	B	Fractions, decimals, percentages & ratios	Apply
32	C	Fractions, decimals, percentages & ratios	Apply
33	A	Fractions, decimals, percentages & ratios	Apply
34	D	Fractions, decimals, percentages & ratios	Understand
35	B	Tree diagrams & multi-stage experiments	Apply
36	C	Tree diagrams & multi-stage experiments	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Tree diagrams & multi-stage experiments	Analyse
38	B	Tree diagrams & multi-stage experiments	Understand
39	D	Tree diagrams & multi-stage experiments	Understand
40	C	Tree diagrams & multi-stage experiments	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 25 (62%) | Remember: 3 (7%) | Understand: 9 (22%)

Ch1**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Place value, reading & writing large numbers	Remember
2	A	Place value, reading & writing large numbers	Remember
3	D	Place value, reading & writing large numbers	Apply
4	B	Place value, reading & writing large numbers	Apply
5	A	Place value, reading & writing large numbers	Remember
6	B	Place value, reading & writing large numbers	Apply
7	C	Place value, reading & writing large numbers	Remember
8	A	Place value, reading & writing large numbers	Apply
9	D	Skip counting (10,000s, 100,000s, 500,000s)	Apply
10	C	Skip counting (10,000s, 100,000s, 500,000s)	Apply
11	B	Skip counting (10,000s, 100,000s, 500,000s)	Apply
12	A	Skip counting (10,000s, 100,000s, 500,000s)	Apply
13	C	Skip counting (10,000s, 100,000s, 500,000s)	Understand
14	D	Comparing & ordering whole numbers	Apply
15	B	Comparing & ordering whole numbers	Apply
16	A	Comparing & ordering whole numbers	Apply
17	D	Comparing & ordering whole numbers	Apply
18	C	Comparing & ordering whole numbers	Apply
19	B	Comparing & ordering whole numbers	Understand
20	D	Standard form	Apply
21	A	Standard form	Apply
22	C	Standard form	Apply
23	B	Standard form	Apply
24	A	Standard form	Remember
25	D	Standard form	Apply
26	C	Standard form	Apply
27	B	Standard form	Analyse
28	A	Standard form	Remember
29	C	Significant figures & decimal places	Apply
30	B	Significant figures & decimal places	Apply
31	D	Significant figures & decimal places	Apply
32	A	Significant figures & decimal places	Apply
33	C	Significant figures & decimal places	Understand
34	B	Significant figures & decimal places	Apply
35	D	Significant figures & decimal places	Apply
36	A	Real-life & word problems	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	C	Real-life & word problems	Apply
38	B	Real-life & word problems	Apply
39	D	Real-life & word problems	Analyse
40	A	Real-life & word problems	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 27 (67%) | Remember: 6 (15%) | Understand: 3 (7%)

Ch2

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Sets of factors & common factors	Remember
2	A	Sets of factors & common factors	Remember
3	D	Sets of factors & common factors	Apply
4	B	Sets of factors & common factors	Apply
5	A	Sets of factors & common factors	Remember
6	B	Sets of factors & common factors	Apply
7	C	Sets of factors & common factors	Analyse
8	A	Identifying perfect squares	Remember
9	D	Identifying perfect squares	Understand
10	C	Identifying perfect squares	Apply
11	B	Identifying perfect squares	Apply
12	A	Identifying perfect squares	Understand
13	C	Identifying perfect squares	Apply
14	D	Identifying perfect squares	Remember
15	B	Square roots	Remember
16	A	Square roots	Apply
17	D	Square roots	Apply
18	C	Square roots	Apply
19	B	Square roots	Apply
20	D	Square roots	Apply
21	A	Square roots	Apply
22	C	Square roots	Apply
23	B	Union & intersection of sets	Remember
24	A	Union & intersection of sets	Understand
25	D	Union & intersection of sets	Apply
26	C	Union & intersection of sets	Apply
27	B	Union & intersection of sets	Understand
28	A	Union & intersection of sets	Remember
29	C	Union & intersection of sets	Apply
30	B	Union & intersection of sets	Apply
31	D	Union & intersection of sets	Understand
32	A	Union & intersection of sets	Apply
33	C	Venn diagrams & real-life problems	Understand
34	B	Venn diagrams & real-life problems	Analyse
35	D	Venn diagrams & real-life problems	Apply
36	A	Venn diagrams & real-life problems	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Venn diagrams & real-life problems	Analyse
38	B	Venn diagrams & real-life problems	Apply
39	D	Venn diagrams & real-life problems	Apply
40	A	Venn diagrams & real-life problems	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 22 (55%) | Remember: 8 (20%) | Understand: 6 (15%)

Ch3

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Multiplying & dividing by powers of 10; be...	Remember
2	A	Multiplying & dividing by powers of 10; be...	Apply
3	D	Multiplying & dividing by powers of 10; be...	Apply
4	B	Multiplying & dividing by powers of 10; be...	Apply
5	A	Multiplying & dividing by powers of 10; be...	Remember
6	B	Multiplying & dividing by powers of 10; be...	Remember
7	C	Multiplying & dividing by powers of 10; be...	Remember
8	A	Mental strategies (halving & doubling, facts)	Understand
9	D	Mental strategies (halving & doubling, facts)	Apply
10	C	Mental strategies (halving & doubling, facts)	Apply
11	B	Mental strategies (halving & doubling, facts)	Remember
12	A	Mental strategies (halving & doubling, facts)	Apply
13	C	Mental strategies (halving & doubling, facts)	Apply
14	D	Addition & subtraction of multi-digit numbers	Apply
15	B	Addition & subtraction of multi-digit numbers	Apply
16	A	Addition & subtraction of multi-digit numbers	Apply
17	D	Addition & subtraction of multi-digit numbers	Apply
18	C	Addition & subtraction of multi-digit numbers	Analyse
19	B	Addition & subtraction of multi-digit numbers	Apply
20	D	Addition & subtraction of multi-digit numbers	Apply
21	A	Addition & subtraction of multi-digit numbers	Apply
22	C	Multiplication & division by 2- and 3-digi...	Apply
23	B	Multiplication & division by 2- and 3-digi...	Apply
24	A	Multiplication & division by 2- and 3-digi...	Apply
25	D	Multiplication & division by 2- and 3-digi...	Apply
26	C	Multiplication & division by 2- and 3-digi...	Apply
27	B	Multiplication & division by 2- and 3-digi...	Apply
28	A	Multiplication & division by 2- and 3-digi...	Understand
29	C	Multiplication & division by 2- and 3-digi...	Apply
30	B	Multiplication & division by 2- and 3-digi...	Analyse
31	D	Multiplication & division by 2- and 3-digi...	Apply
32	A	Story problems (decimals, four operations)	Apply
33	C	Story problems (decimals, four operations)	Apply
34	B	Story problems (decimals, four operations)	Apply
35	D	Story problems (decimals, four operations)	Apply
36	A	Story problems (decimals, four operations)	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Story problems (decimals, four operations)	Apply
38	B	Story problems (decimals, four operations)	Apply
39	D	Story problems (decimals, four operations)	Apply
40	A	Story problems (decimals, four operations)	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 30 (75%) | Remember: 5 (12%) | Understand: 2 (5%)

Ch4

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Laws of indices (identify & explain)	Remember
2	A	Laws of indices (identify & explain)	Remember
3	D	Laws of indices (identify & explain)	Remember
4	B	Laws of indices (identify & explain)	Remember
5	A	Laws of indices (identify & explain)	Remember
6	B	Laws of indices (identify & explain)	Understand
7	C	Laws of indices (identify & explain)	Understand
8	A	Laws of indices (identify & explain)	Remember
9	D	Simplifying & evaluating powers (PEDMAS)	Apply
10	C	Simplifying & evaluating powers (PEDMAS)	Apply
11	B	Simplifying & evaluating powers (PEDMAS)	Apply
12	A	Simplifying & evaluating powers (PEDMAS)	Apply
13	C	Simplifying & evaluating powers (PEDMAS)	Apply
14	B	Simplifying & evaluating powers (PEDMAS)	Analyse
15	D	Simplifying & evaluating powers (PEDMAS)	Apply
16	A	Simplifying & evaluating powers (PEDMAS)	Apply
17	C	Simplifying & evaluating powers (PEDMAS)	Apply
18	B	Simplifying & evaluating powers (PEDMAS)	Apply
19	D	Simplifying & evaluating powers (PEDMAS)	Apply
20	A	Simplifying & evaluating powers (PEDMAS)	Apply
21	C	Exponential equations	Apply
22	B	Exponential equations	Apply
23	D	Exponential equations	Apply
24	A	Exponential equations	Apply
25	C	Exponential equations	Apply
26	B	Exponential equations	Apply
27	D	Exponential equations	Analyse
28	A	Exponential equations	Analyse
29	C	Exponential equations	Apply
30	B	Exponential equations	Apply
31	D	Real-life problems (powers of natural numb...	Apply
32	A	Real-life problems (powers of natural numb...	Analyse
33	C	Real-life problems (powers of natural numb...	Understand
34	B	Real-life problems (powers of natural numb...	Apply
35	D	Real-life problems (powers of natural numb...	Apply
36	A	Real-life problems (powers of natural numb...	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Real-life problems (powers of natural numb...	Apply
38	B	Real-life problems (powers of natural numb...	Analyse
39	D	Real-life problems (powers of natural numb...	Apply
40	A	Real-life problems (powers of natural numb...	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 25 (62%) | Remember: 6 (15%) | Understand: 3 (7%)

Ch5

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Review of fractions (equivalent, simplest,...	Apply
2	A	Review of fractions (equivalent, simplest,...	Apply
3	C	Review of fractions (equivalent, simplest,...	Apply
4	B	Review of fractions (equivalent, simplest,...	Apply
5	D	Review of fractions (equivalent, simplest,...	Apply
6	A	Review of fractions (equivalent, simplest,...	Analyse
7	C	Review of fractions (equivalent, simplest,...	Apply
8	B	Review of fractions (equivalent, simplest,...	Understand
9	D	Four operations on fractions	Apply
10	A	Four operations on fractions	Apply
11	C	Four operations on fractions	Apply
12	B	Four operations on fractions	Apply
13	D	Four operations on fractions	Apply
14	A	Four operations on fractions	Apply
15	C	Four operations on fractions	Apply
16	B	Four operations on fractions	Apply
17	D	Four operations on fractions	Remember
18	A	Four operations on fractions	Apply
19	C	Order of operations (BODMAS) with fractions	Analyse
20	B	Order of operations (BODMAS) with fractions	Analyse
21	A	Order of operations (BODMAS) with fractions	Analyse
22	D	Order of operations (BODMAS) with fractions	Apply
23	C	Order of operations (BODMAS) with fractions	Apply
24	B	Order of operations (BODMAS) with fractions	Apply
25	A	Order of operations (BODMAS) with fractions	Apply
26	C	Order of operations (BODMAS) with fractions	Apply
27	B	Word problems & fractions of quantities	Apply
28	A	Word problems & fractions of quantities	Apply
29	D	Word problems & fractions of quantities	Analyse
30	C	Word problems & fractions of quantities	Apply
31	B	Word problems & fractions of quantities	Apply
32	C	Word problems & fractions of quantities	Apply
33	A	Word problems & fractions of quantities	Apply
34	D	Word problems & fractions of quantities	Understand
35	B	Word problems & fractions of quantities	Apply
36	A	Word problems & fractions of quantities	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Word problems & fractions of quantities	Apply
38	B	Word problems & fractions of quantities	Apply
39	D	Word problems & fractions of quantities	Analyse
40	A	Word problems & fractions of quantities	Understand

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 30 (75%) | Remember: 1 (2%) | Understand: 3 (7%)

Ch6

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	Unit conversion by ratio reasoning	Remember
2	A	Unit conversion by ratio reasoning	Apply
3	C	Unit conversion by ratio reasoning	Apply
4	B	Unit conversion by ratio reasoning	Apply
5	D	Unit conversion by ratio reasoning	Understand
6	A	Unit conversion by ratio reasoning	Apply
7	C	Unit conversion by ratio reasoning	Analyse
8	B	Unit conversion by ratio reasoning	Analyse
9	D	Unit rates, unit pricing & constant speed	Apply
10	A	Unit rates, unit pricing & constant speed	Apply
11	C	Unit rates, unit pricing & constant speed	Apply
12	B	Unit rates, unit pricing & constant speed	Apply
13	D	Unit rates, unit pricing & constant speed	Apply
14	A	Unit rates, unit pricing & constant speed	Apply
15	C	Unit rates, unit pricing & constant speed	Apply
16	B	Unit rates, unit pricing & constant speed	Apply
17	D	Unit rates, unit pricing & constant speed	Apply
18	A	Unit rates, unit pricing & constant speed	Analyse
19	C	Speed & distance-time graphs	Understand
20	B	Speed & distance-time graphs	Understand
21	A	Speed & distance-time graphs	Apply
22	D	Speed & distance-time graphs	Apply
23	C	Speed & distance-time graphs	Understand
24	B	Speed & distance-time graphs	Apply
25	A	Speed & distance-time graphs	Apply
26	C	Speed & distance-time graphs	Apply
27	B	Proportional relationships	Understand
28	A	Proportional relationships	Apply
29	D	Proportional relationships	Understand
30	C	Proportional relationships	Understand
31	B	Proportional relationships	Understand
32	C	Proportional relationships	Apply
33	A	Proportional relationships	Apply
34	D	Constant of proportionality	Analyse
35	B	Constant of proportionality	Analyse
36	A	Constant of proportionality	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	C	Constant of proportionality	Apply
38	B	Constant of proportionality	Remember
39	D	Constant of proportionality	Apply
40	A	Constant of proportionality	Apply

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 10 (25%) | D: 9 (22%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 25 (62%) | Remember: 2 (5%) | Understand: 8 (20%)

Ch7

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	Gradient concept & formula	Remember
2	D	Gradient concept & formula	Remember
3	A	Gradient concept & formula	Understand
4	C	Gradient concept & formula	Understand
5	B	Gradient concept & formula	Remember
6	D	Gradient concept & formula	Remember
7	A	Gradient from coordinates	Apply
8	C	Gradient from coordinates	Apply
9	B	Gradient from coordinates	Apply
10	D	Gradient from coordinates	Apply
11	A	Gradient from coordinates	Analyse
12	C	Gradient from coordinates	Analyse
13	B	Equations of straight lines ($y = mx + c$)	Apply
14	D	Equations of straight lines ($y = mx + c$)	Apply
15	A	Equations of straight lines ($y = mx + c$)	Apply
16	C	Equations of straight lines ($y = mx + c$)	Apply
17	B	Equations of straight lines ($y = mx + c$)	Apply
18	D	Equations of straight lines ($y = mx + c$)	Apply
19	A	Equations of straight lines ($y = mx + c$)	Apply
20	C	Equations of straight lines ($y = mx + c$)	Apply
21	B	Equations of straight lines ($y = mx + c$)	Analyse
22	D	Equations of straight lines ($y = mx + c$)	Apply
23	A	Equations of straight lines ($y = mx + c$)	Apply
24	C	Equations of straight lines ($y = mx + c$)	Understand
25	A	Tables of values & reading graphs	Apply
26	B	Tables of values & reading graphs	Apply
27	C	Tables of values & reading graphs	Apply
28	D	Tables of values & reading graphs	Apply
29	A	Tables of values & reading graphs	Remember
30	B	Tables of values & reading graphs	Apply
31	C	Tables of values & reading graphs	Analyse
32	D	Tables of values & reading graphs	Apply
33	A	Real-life linear graphs	Apply
34	C	Real-life linear graphs	Apply
35	B	Real-life linear graphs	Apply
36	D	Real-life linear graphs	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Real-life linear graphs	Apply
38	C	Real-life linear graphs	Apply
39	B	Real-life linear graphs	Apply
40	D	Real-life linear graphs	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 28 (70%) | Remember: 5 (12%) | Understand: 3 (7%)

Ch8

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	Expanding brackets (distributive property)	Apply
2	B	Expanding brackets (distributive property)	Apply
3	D	Expanding brackets (distributive property)	Apply
4	A	Expanding brackets (distributive property)	Apply
5	B	Expanding brackets (distributive property)	Apply
6	A	Expanding brackets (distributive property)	Apply
7	D	Expanding brackets (distributive property)	Apply
8	C	Expanding brackets (distributive property)	Apply
9	B	Multiplication of binomial expressions	Apply
10	A	Multiplication of binomial expressions	Apply
11	D	Multiplication of binomial expressions	Apply
12	C	Multiplication of binomial expressions	Apply
13	B	Multiplication of binomial expressions	Apply
14	A	Multiplication of binomial expressions	Apply
15	D	Multiplication of binomial expressions	Apply
16	B	Multiplication of binomial expressions	Understand
17	A	Operations on algebraic fractions	Apply
18	C	Operations on algebraic fractions	Apply
19	D	Operations on algebraic fractions	Analyse
20	B	Operations on algebraic fractions	Apply
21	A	Operations on algebraic fractions	Apply
22	C	Operations on algebraic fractions	Apply
23	B	Operations on algebraic fractions	Apply
24	D	Operations on algebraic fractions	Analyse
25	A	Substitution & evaluation	Apply
26	D	Substitution & evaluation	Apply
27	B	Substitution & evaluation	Apply
28	C	Substitution & evaluation	Apply
29	A	Substitution & evaluation	Apply
30	D	Substitution & evaluation	Apply
31	C	Substitution & evaluation	Analyse
32	A	Substitution & evaluation	Apply
33	D	Factorisation	Apply
34	B	Factorisation	Apply
35	C	Factorisation	Apply
36	B	Factorisation	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Factorisation	Apply
38	C	Factorisation	Apply
39	D	Factorisation	Analyse
40	B	Factorisation	Analyse

Answer Distribution:

A: 10 (25%) | B: 11 (27%) | C: 9 (22%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 34 (85%) | Understand: 1 (2%)

Ch9**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Translating word problems into inequalities	Apply
2	C	Translating word problems into inequalities	Apply
3	B	Translating word problems into inequalities	Analyse
4	D	Translating word problems into inequalities	Apply
5	A	Translating word problems into inequalities	Apply
6	C	Translating word problems into inequalities	Understand
7	B	Translating word problems into inequalities	Apply
8	D	Translating word problems into inequalities	Understand
9	A	Solving simple linear inequalities	Apply
10	D	Solving simple linear inequalities	Apply
11	C	Solving simple linear inequalities	Apply
12	B	Solving simple linear inequalities	Apply
13	A	Solving simple linear inequalities	Apply
14	D	Solving simple linear inequalities	Apply
15	B	Solving simple linear inequalities	Apply
16	A	Solving simple linear inequalities	Analyse
17	C	Solving simple linear inequalities	Analyse
18	B	Solving simple linear inequalities	Apply
19	D	Solving simple linear inequalities	Apply
20	A	Solving simple linear inequalities	Apply
21	C	Solving simple linear inequalities	Apply
22	B	Solving simple linear inequalities	Analyse
23	D	Solution sets in given domains	Apply
24	A	Solution sets in given domains	Apply
25	B	Solution sets in given domains	Apply
26	C	Solution sets in given domains	Apply
27	A	Solution sets in given domains	Apply
28	D	Solution sets in given domains	Understand
29	B	Solution sets in given domains	Apply
30	C	Solution sets in given domains	Analyse
31	A	Solution sets in given domains	Apply
32	D	Solution sets in given domains	Analyse
33	B	Number lines & real-life applications	Understand
34	A	Number lines & real-life applications	Understand
35	C	Number lines & real-life applications	Apply
36	D	Number lines & real-life applications	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	Number lines & real-life applications	Apply
38	C	Number lines & real-life applications	Apply
39	B	Number lines & real-life applications	Apply
40	D	Number lines & real-life applications	Analyse

Answer Distribution:

A: 11 (27%) | B: 10 (25%) | C: 9 (22%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 7 (17%) | Apply: 28 (70%) | Understand: 5 (12%)

Mathematics Formula Sheet — Basic 8

Numbers and Operations

Directed Numbers

Positive and negative numbers on the number line

Adding Negatives

$$a + (-b) = a - b$$

Subtracting Negatives

$$a - (-b) = a + b$$

Multiplying Signs

$$(+)\times(+) = +, (-)\times(-) = +, (+)\times(-) = -, (-)\times(+) = -$$

Index Notation

$$a^n = a \times a \times \dots \text{ (n times)}$$

Standard Form

$$a \times 10^n \text{ where } 1 \leq a < 10$$

Ratio

$a : b$ (comparison of quantities)

Proportion

$$a/b = c/d \text{ (equivalent ratios)}$$

Direct Proportion

$$y = kx \text{ (y increases as x increases)}$$

Inverse Proportion

$$y = k/x \text{ (y decreases as x increases)}$$

Rate

Comparison of two different quantities

Algebra

Simplify Expressions

$$\text{Combine like terms: } 2x + 3x = 5x$$

Expand Brackets

$$a(b + c) = ab + ac$$

Factorise

$$ab + ac = a(b + c)$$

Linear Equation

$$ax + b = c$$

Solve Linear Equation

$$x = (c - b) / a$$

Simultaneous Equations

Two equations with two unknowns

Inequality

$$a < b, a > b, a \leq b, a \geq b$$

Sequence (nth term)

$$a_n = a + (n-1)d \text{ for arithmetic sequences}$$

Common Difference

$$d = a_2 - a_1$$

Geometry

Pythagoras' Theorem

$$a^2 + b^2 = c^2 \text{ (c is hypotenuse)}$$

Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint Formula

$$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Gradient of Line

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Equation of Line

$$y = mx + c$$

Volume of Cylinder

$$V = \pi r^2 h$$

Volume of Cone

$$V = \frac{1}{3} \pi r^2 h$$

Volume of Sphere

$$V = \frac{4}{3} \pi r^3$$

Surface Area of Cylinder	$SA = 2\pi r^2 + 2\pi rh$
Surface Area of Sphere	$SA = 4\pi r^2$
Interior Angle (Regular Polygon)	Each angle = $(n-2) \times 180^\circ / n$
Exterior Angle (Regular Polygon)	Each angle = $360^\circ / n$
Sum of Interior Angles	Sum = $(n-2) \times 180^\circ$
Transformations	Translation, Rotation, Reflection, Enlargement

Trigonometry

SOH	$\sin \theta = \text{Opposite} / \text{Hypotenuse}$
CAH	$\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
TOA	$\tan \theta = \text{Opposite} / \text{Adjacent}$
Find Angle	$\theta = \sin^{-1}(O/H), \theta = \cos^{-1}(A/H), \theta = \tan^{-1}(O/A)$

Statistics and Probability

Mean of Grouped Data	$\text{Mean} = \Sigma(fx) / \Sigma f$
Modal Class	Class with highest frequency
Median Class	Class containing the $(n/2)$ th value
Probability	$P(A) = n(A) / n(S)$
Complement	$P(A') = 1 - P(A)$
Combined Events	$P(A \text{ and } B) = P(A) \times P(B)$ for independent events

Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: www.ges.gov.gh

NaCCA: www.nacca.gov.gh

WAEC: www.waec.org

About Beacon Educational Consult

Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

Our mission is to support students in achieving academic excellence through comprehensive, curriculum-aligned resources that prepare them for the Basic Education Certificate Examination (BECE) and beyond.

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